Systems and Complexity Science: The Generation, Persistence, and Evolution of Order — The Governing Principles of Open Complex Giant Systems

A Formal Unified Monograph: From First Principles Qualitative Eight Treatises, Dual-Helix Five-Dimensional Manifold, to Metacognitive Re-boundary Creation

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This axiomatic system does not originate from armchair deduction, but is the fruit of twenty-two years of practical accumulation and scientific transcendence.

From 2004 to 2007 was my early accumulation period in enterprise information systems and data structures. During this period, I clarified the structure of element inventories, causal relationships, and underlying constraints, laying the foundation for dealing with more complex planning and scheduling.

From 2007 to 2020, I worked at Lenovo Group. Over these thirteen years, our team built an end-to-end closed-loop autonomous decision-making supply chain planning system—IPS ("Integrated Planning Solution")—in the world's most complex discrete manufacturing battlefield.

This system is an "industrial autonomous driving" engine. From delivery promise, multi-level kitting absorption, production scheduling linkage, assembly line pulling to warehouse dispatching, the entire chain achieved "human-out-of-the-loop, autonomous decision-making." In the global manufacturing network, including LCFC Hefei (a World Economic Forum "Lighthouse Factory"), its operational efficiency under scheduling has been strictly verified by official operational data: the delivery response rate rose from 54% to 98%, order delivery accuracy increased by 32%, and overall inventory turnover increased by 1.9 times, directly releasing billions of RMB in liquidity.

This is one of the very few engineering empirical proofs to date where humanity has achieved a "computable closed loop" when facing complexity science and the governance of open complex giant systems. On a physical battlefield of factorial complexity, it achieved self-healing closed-loop operation at millisecond decision speeds.

Engineering success prompted the pursuit of its essence, driving the exploration to transcend from engineering implementation to theoretical abstraction. Global software giants attempted to solve highly non-linear dynamic networks using open-loop, fragmented modular architectures, which is physically doomed to fail in finding global optimality.

The path we forged proved through empirical engineering the power of Qian Xuesen's "Metasynthetic Method" and "human-machine synergy" in a hundred-billion-scale physical world. Precisely because closed-loop computability was verified on the physical battlefield, I was able to stand back at the origin, shedding all burdens of post-hoc academic jargon. Starting from the most plain boundary creation and disorder, following the shortest logical path, I recognized the Second Law of Thermodynamics, Shannon Information Theory, and Ashby's Law of Requisite Variety as isomorphic

mappings, and under prior boundary creation conditions, proved intra-boundary closed-loop computability and extreme uniqueness.

Stepping back to re-examine the whole edifice, I realized that *Value Chain Physics* is precisely the physical projection of systems and complexity science in supply chain management; and since a projection exists, universal physical and logical isomorphisms across domains must inevitably be extracted.

I took Occam's razor to shave off all redundant jargon and intermediate assumptions accumulated over the past twenty-two years. I returned to the irreducible logical starting point—starting from the most plain and everyday "disorder" and "partitioning", adhering to the Seven Iron Laws of "no overdraft, shortest path, parsimony of words, empirical authority, respect before assimilation, constitution first, and falsifiability", forging the entire system into this rigorous, rock-solid minimal axiomatic text of *The Origin of Order*.

The Seven Laws of this book are listed below:

1. **First Law (No Overdraft)**: Every new term must strictly depend on previously defined terms to be born. Terms grow like trees from seed sprouting to branching leaves, without sudden mutation. Cross-volume usage of undefined terminology is strictly prohibited.
2. **Second Law (Shortest Path)**: From the most primitive axiom to the most realistic inference, there must be no logical leaps in between. Every step must stand firmly on the shoulders of the previous step.
3. **Third Law (Parsimony of Words)**: Use the most plain, everyday, self-evident words possible. Entities should not be multiplied without necessity. Wherever "boundary partitioning" can be used, do not use "modeling"; wherever "clarity of observation" can be used, do not use "metacognitive function."
4. **Fourth Law (Empirical Authority)**: The end of the logic chain must connect to real-world engineering validation. Use the stable operation of a hundred-billion-scale manufacturing network to provide the most indisputable decisive test for the entire axiomatic system.
5. **Fifth Law (Respect Before Assimilation)**: Explain predecessors using axioms, rather than proving oneself using predecessors. Ancient wisdom is concentrated in the end-of-volume special chapter on isomorphism. Let every fragment of ancient wisdom find its place in your complete puzzle.
6. **Sixth Law (Constitution First, Continuation Second)**: A complete qualitative axiomatic system must first be established using the three laws before using this system to re-explain predecessors.
7. **Seventh Law (Falsifiability)**: The entire book must define explicit falsifiability deadlines—order persisting without partitioning, order not dissipating without constraints, order evolving with zero residuals.

Practice precedes theory, and empirical proof precedes axioms.

The mission of this book is not only to establish universal physical and logical specifications for data structures and algorithms in complex systems, but also, with the greatest respect, to carry forward the spark of exploration of Eastern and Western sages and scientists over thousands of years.

Explain predecessors using axioms, rather than proving oneself using predecessors. From Spencer-Brown's distinction, Kant's legislation, and Popper's falsifiability, to Qian Xuesen's open complex giant systems, Wu Xuemou's Pansystems Theory, and Cao Longbing's Non-IID theory—the formal closed loop established by this system does not abandon predecessors, but rather allows every fragment of wisdom from ancient sages to find its ultimate position in the complete physical puzzle.

Purifying the source and bridging past and future. We will start from the most primitive plain axioms, seamlessly penetrating to hard-core formal mathematics and engineering codes, attempting to

continue and answer the unfinished ultimate questions of ancient sages in formal and engineering dimensions.

Introduction: The Origin of Order and Metacognitive Algorithms

Human civilization's exploration of the world over thousands of years is essentially the process of an observer establishing order through boundary creation.

Humans cannot physically access the objective actual reality directly, fully, and losslessly; we can only observe, simulate, and govern the real physical world by establishing an "ought-to-be system" through observer boundary creation. And the essence of an ought-to-be system is precisely "order."

Empiricism is the ought-to-be established through intuitive trial and error; philosophy is the ought-to-be established through second-order reflection; physics is the ought-to-be established through microscopic rigid rules; systems science is the ought-to-be established through macroscopic collaborative topology. Empiricism, philosophy, physics, and systems science are all cornerstones of human order construction. Systems and Complexity Science, as the horizontal axis studying the universal laws of generation, persistence, and evolution of ought-to-be systems across all specific disciplines, naturally occupies the common underlying mediating position of all specific disciplines, and is therefore established as "Metascience (i.e., the Science of Sciences)".

However, facing open complex giant systems with factorial complexity, humanity encounters the most difficult challenge of ought-to-be governance. A single microscopic disassembly or open-loop description cannot accomplish computable dimension-reduced governance. This book, *Systems and Complexity Science: The Generation, Persistence, and Evolution of Order*, steps back to the most primitive logical starting point, uses Occam's razor to shave off all intermediate redundancies, and establishes the **first principles of order generation, persistence, and evolution** at the dimension of the **underlying meta-algorithm**. This meta-algorithm is essentially a self-consistent closed loop where an observer sets ought-to-be by boundary creation, sets rigid rules by ought-to-be, inspects residuals by rules, and drives metacognitive re-boundary creation by residuals. It governs concrete calculation rules inside the system, constituting the living mechanism of the system.

Once the order of the most difficult open complex giant systems is established and closed-loop computability is achieved, empirical trial and error, philosophical self-reflection, institutional norms, and reductionist models all transform into their boundary-creation dimension reduction and degenerate fallbacks in local phase space. It seamlessly fuses the intuition of empiricism, the self-reflection of philosophy, the constraints of physics, and the topology of systems into a formal closed loop of "generation by boundary creation, persistence by rigid constraints, and evolution by residual feedback." Far from discarding predecessors, it provides universal executable physical and mathematical source codes for governing open complex giant systems.

Volume I: Order

— The Origin of Order: The First Principles of Open Complex Giant Systems

Chapter 1: Definitions and Axioms

1.1 Observer and Boundary Creation (Definition 1)

Without an observer's boundary creation, there is no ought-to-be representational world (Ω). Once disorder is observed, it immediately manifests as the world. Above physics, the world is order.

- **Definition 1 (Disorder)**: The whole unpartitioned by an observer is called disorder. Disorder is not nothingness, but the original nature of all that has not been partitioned.
- **Definition 2 (Observer, Cognition, Knowledge)**: In disorder, whatever can distinguish inside from outside is an observer. Cognition (Zhi) is the clarity of observation, the source of boundary creation; Knowledge (Shi) is the power of construction, the boundary of order establishment. Cognition and Knowledge are co-existent and co-emergent.
- **Axiom 0 (Axiom Zero · Observer Emergence Law)**: The observer is not an a priori preset starting point, but a self-sustaining steady state naturally reached within disorder. Without an observer's boundary creation, there is no ought-to-be representational world (Ω).
    - **Deduction**: If there is no self-sustaining steady state inside disorder, all physical structures and observation activities collapse into dissolution. But disorder itself is not nothingness; local self-sustaining steady states must form in its spontaneous fluctuations and phase transitions. This self-sustaining steady state is the ontogenetic starting point for observer and boundary creation.
- **Definition 3 (Boundary Creation and Order Establishment)**: The action of an observer partitioning limits using cognition is called boundary creation; the action of establishing rules inside the boundary using knowledge is called order establishment. Inside the boundary is the ought-to-be world (the world is the general term for the ought-to-be inside the boundary), and outside is the disorder background. Rules established by knowledge that cannot be violated are called hard rules.
- **Proposition 1 (Manifestation of the World)**: Without the boundary-creating presence and awareness of an observer, order has nowhere to attach; without boundary creation and order establishment, the ought-to-be representational world (Ω) cannot be established.
    - **Deduction**: According to Definition 1, disorder is the unpartitioned whole. According to Definition 3, the world is the manifestation of the bounded orderly scope after boundary creation. The instant the observer performs boundary creation to cut inside from outside, it not only establishes the ought-to-be boundary, but also simultaneously establishes the all-domain scalar potential energy field and negative entropy attractor pointing to the self-sustaining steady state inside the boundary. If no observer performs boundary creation, the distinction between inside and outside ceases to exist, and states return to unpartitioned disorder. Therefore, observer presence and execution of boundary creation are the prerequisite necessary and sufficient conditions for the manifestation of the world, potential energy fields, and order.

1.2 Actual Reality, Ought-to-be Reality, and Residuals (Definition 2)

When an observer performs boundary creation and observation, three-fold reality simultaneously manifests:

- **Actual Reality**: The infinite background silently occurring in boundary creation. Whether inside or outside the boundary, actual reality fills it, but the observer cannot directly know the full picture of actual reality apart from perceptual media.
- **Ought-to-be Reality**: Order inside the boundary. The local orderly framework constructed by the observer through boundary creation. All that is expressed and measured belongs to ought-to-be. Without an ought-to-be framework, order cannot attach, and everything inside the boundary collapses.
- **Residuals**: Difference computation interfaces reserved and established by the observer in the ought-to-be framework. Regardless of whether the residual numerical value is zero, residuals are monitored in real time by the observer. When deviation occurs between new and old frameworks, residuals are triggered, becoming the sole evolutionary driving force breaking dogma and promoting reconstruction; without residual feedback, order degenerates into rigidity and stagnation.
- **Physical Exclusivity and Intra-boundary Constraints**: Entities inside the ought-to-be framework possess original physical exclusivity (entities in the world cannot occupy overlapping positions) and rule constraints; occupation at one location means obstruction at another, which is the physical foundation of intra-boundary order persistence and hard collision. Intra-boundary entities possess exclusivity; without rule constraints, they inevitably dissipate due to mutual confrontation.

[Geometric Projection Declaration of "One Boundary, Three Facets"]: Actual reality, ought-to-be reality, and residuals simultaneously manifest. Their expansion in mathematics and dynamics inevitably requires corresponding geometric phase spaces and state manifolds to be synchronously present. Therefore, the following terms are not post-hoc tools, but indispensable geometric skeletons when the three-fold reality manifests:

[Terminology Native Growth Chain I: Phase Space (Ξ and Ω), Physical Five-Dimensional Basis, Manifolds, and Norm Differences]

All formal and cybernetic terms in this system are not borrowed from external sources, but strictly grow natively from the primitive action of "boundary creation":

1. **Boundary Creation ⟹ Phase Space Notation Unified Definition (Ξ and Ω)**: Observer performs boundary creation, and entities immediately manifest inside the boundary. Entities are not static, each having its variable phase states. The unpartitioned infinite disorder background is physically called the "unpartitioned full phase space Ξ"; the local finite orderly state set cropped and confirmed by the observer from Ξ after boundary creation is called the "intra-boundary ought-to-be state space $\Omega \subset \Xi$". The dimension-reduction work formula of the boundary creation operator Π is to crop the infinite background into a finite intra-boundary space: $\Pi: \Xi \to \Omega$.
2. **Phase Space ⟹ Minimal Complete Physical Five-Dimensional Basis**: Observer establishes intra-boundary ought-to-be phase space Ω. To describe entity motion and evolution in physical spatiotemporal axes, it necessary and uniquely requires five minimal orthogonal bases—who is moving (Nodes $\mathcal{N}$), who is connected to whom (Topology $\mathcal{T}$), where active boundaries lie (Constraint Clusters $\mathcal{C}$), where current position is (State Vector X), and how it changes next (State Transitions with causal pointers and Residuals ΔX). Fully possessing all five constitutes the minimal complete computable topological manifold $\mathcal{M}_{\text{model}} = \langle \mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X \rangle$ of the ought-to-be phase space in the physical world. (Plain words: every point on this continuous state space composed of these five elements can be smoothly

located according to its neighbors without tearing or breaking. Nodes fix objects, topology fixes connections, constraints fix boundaries, states fix markers, transitions fix steps—fully possessing all five enables smooth evolution in this space).

3. **Hard Rules ⟹ Constraint Trajectories / Rigid Manifold (Manifold, $\Pi_{\perp}$)**: Phase space contains all-encompassing possible states, the vast majority of which are excluded for violating rules. **[Plain Explanation]**: Figuratively speaking, just as trains cannot run wild across the landscape and must strictly travel along iron rails, the vast majority of tangled states in complex system phase space are illegal states violating rules. The continuous legal trajectory channels strictly permitted and smoothly navigable by rigid hard rules in differential geometry are vividly called [Manifolds]; the channel in this system that limits healthy system persistence is specially termed [Rigid Trajectory Manifold $\Pi_{\perp} \subset \Omega$].
4. **Prospective Expectations ⟹ Gap ⟹ Residual ⟹ Norm Difference (Δ)**: The ought-to-be framework not only describes the present, but inevitably projects and pre-judges actual reality states in the next temporal step (prospective expectation). When actual reality manifests next time, the indelible deviation and gap between objective actual reality and subjective expectation is the [Residual]. The mathematical formula measuring the overall absolute intensity of this gap is the [State Norm Difference $\Delta = \| \Omega_t - \Omega_{t-1} \|$].

1.3 The Three Axioms of Order Generation, Persistence, and Evolution

- **Axiom 1 (First Law · Generation Axiom / Collapse Without Ought-to-be)**: If the observer does not perform boundary creation to construct an ought-to-be framework, nothing in chaos can persist. Without an ought-to-be framework, everything collapses into dissolution.
- **Axiom 2 (Second Law · Persistence Axiom / Dissipation Without Constraints)**: Observer enters the world by boundary creation establishing ought-to-be; if self-consistent rules and rigid constraints are not formulated, elements inside the ought-to-be framework inevitably dissipate due to disorderly confrontation. Without rigid constraints and rules, order inside the boundary inevitably dissipates.
- **Axiom 3 (Third Law · Evolution Axiom / Stagnation Without Residuals)**: The ought-to-be framework inevitably possesses local partiality when facing infinite actual reality. If the observer does not identify ought-to-be residuals, the framework solidifies into dogma, and evolution stagnates. Without residual feedback, generational evolution stagnates.

[Terminology Native Growth Chain II: Survival Baseline, Prior, Compact Support, and Sub-Gaussian]

1. **Survival Baseline ⟹ Top-Level Prior (Bayesian Prior) ⟹ Compact Support Operator (E_{supp}) and Sub-Gaussian Truncation**: Observer constructs ought-to-be system primarily to preserve own survival, thus inevitably pre-setting above all specific rules an absolute baseline "rather shatter all local indicators than cross", called in cognitive science [Top-Level Prior / Conscience]. **[Plain Explanation]**: Figuratively speaking, no matter how much a person pursues commercial profit, there is always a red line in the heart "rather give up business than violate law and conscience"; before conducting specific indicator optimization, this uncrossable ultimate anchor stone pre-set in the heart of the system is called [Top-Level Prior (Bayesian Prior)] in probabilistic cognitive science. Bayesian refers to the reasoning method where observers update judgment based on existing beliefs when encountering new evidence. Top-level prior is the unshakeable highest anchor point in this reasoning. When local optimization expands infinitely causing residual violent divergence, this baseline **forcibly locks the active territory of residuals inside a safe dead zone** at the physical critical point, called in mathematics [Compact Support (E_{supp})]; risk probability states whose infinitely divergent tails are cut off by compact support converge statistically to [Sub-Gaussian] distributions. Qualitatively speaking, the essence of sub-Gaussian truncation is for the

observer, relying on uncrossable baseline conscience, to forcibly sever heavy-tailed infinite divergence of blind profit-seeking at physical critical points, firmly anchoring the system inside a compact territory of safe survival.

Chapter 2: The Meta-Theorem of Order and Dynamic Essence

2.1 The Meta-Theorem of Order (Necessary and Sufficient Condition Loop)

For order to persist and evolve generation by generation, if and only if it simultaneously satisfies the following three necessary and sufficient conditions:

$$\boldsymbol{Meta\text{-}Theorem\ of\ Order} \Leftrightarrow \begin{cases} \boldsymbol{Build\ Boundary\ Framework} \Longrightarrow \boldsymbol{Prevent\ Collapse} \\ \boldsymbol{Define\ Rigid\ Constraints} \Longrightarrow \boldsymbol{Prevent\ Dissipation} \\ \boldsymbol{Identify\ Residual\ Feedback} \Longrightarrow \boldsymbol{Prevent\ Stagnation} \end{cases}$$

The three are not sequential, but three facets of one boundary, simultaneously present; missing one collapses the boundary; fully possessing all three enables endless growth.

2.2 Dynamic Essence of Order and Human Constructs

Order is never a static objective entity.

The physical and philosophical essence of order: the dynamic closed-loop process where the observer creates boundary to give birth to order, formulates rules and constraints to persist order, and identifies residuals to evolve order.

This closed loop is the ontogenetic starting point of all human constructs. Whatever humans construct—whether intuitive common sense of empiricism, hypotheses of scientific paradigms, institutional norms, or rhythms of artistic forms—without exception obeys boundary creation establishing ought-to-be, rigid rules establishing constraints, and residual-driven evolution. Apart from these three, no order in human civilization can be born or sustained (**Boundary Creation Relativity Law**: living is absolute, optimality is relative inside boundary creation).

2.3 Natural Birth of Logic, Models, and Algorithms

Once the ought-to-be framework is established, the observer, to maintain internal self-consistency of rules, inevitably derives the skeleton and tools of evolution:

- **Logic**: The skeleton maintaining internal self-consistency and contradiction-free deduction inside the ought-to-be framework is called logic.
- **Model**: The structured expression of the ought-to-be framework in observer cognition is called a model. Models establish static projections of entities and rule boundaries inside the boundary.
- **Algorithm**: The sequential deduction steps of rules inside the boundary along the time axis are called algorithms.

Deduction Conclusion: Models are static projections of ought-to-be, algorithms are dynamic evolutions of ought-to-be, and logic is the bond of self-consistency between the two.

[Terminology Native Growth Chain III: First-Order Calculus, Gödel Deadlock, Metacognitive Constitutional Amendment Operator, and Tensor Product]

1. **First-Order Calculus ⟹ Self-Referential Exhaustion and Gödel Deadlock / Turing Halting**: Step evolution where observers mechanically execute preset rules inside established boundaries is called [First-Order Calculus]. When system encounters major external impact not mapped into original framework, if it refuses to re-partition boundaries and foolishly attempts to prove or eliminate this residual within old rules, old rules inevitably fall into infinite loops of "explaining itself using own definition". This physical halting state of **"unable to self-prove or self-heal relying on intra-boundary rules"** is mathematically equivalent to [Gödel Incompleteness and Turing Halting Deadlock] (Gödel incompleteness and Turing halting are actually two statements of the same physical halting state in two disciplines: former says "system cannot self-prove inside old rules", latter says "algorithm cannot halt inside old rules". The two are isomorphic, both referring to first-order calculus failing to self-heal inside old boundaries).
2. **Re-boundary Creation ⟹ Second-Order Metacognitive Constitutional Amendment Operator** Φ: Facing first-order deadlock, all first-order calculus fails. The system's only way out is jumping out of current rules, re-examining, breaking old boundaries, and drawing new boundaries containing new dimensions, an action called [Re-boundary Creation]. The formal work formula describing leap rewriting from "old boundary" to "new boundary" is the [Second-Order Metacognitive Constitutional Amendment Operator $\Phi: \Pi_k \to \Pi_{k+1}$].
3. **Unity of Essence and Function & Orthogonal Fusion ⟹ Tensor Product ($\otimes$)**: Governance of open complex giant systems requires two forces of work: human brain responsible for "jumping out of deadlock, re-drawing big boundary" (second-order function), and machine responsible for "extremely fast concurrent computation inside set big boundary" (first-order essence). These two forces expand in independent directions and can never replace each other (orthogonal), but in every genuine system work, they must be tightly interlocked and completely entangled together (unity). This mathematical relationship of "expanding independently yet indivisibly fusing into one to perform work" is the [Tensor Product $\otimes$].

Chapter 3: Falsifiability Boundaries

3.1 Falsifiability Deadlines of the System

This system declares its own falsifiability deadlines. If any counterexample below is captured in real observation or logical deduction, this theory collapses:

1. **[Epistemological / Emergence Level Deadline] No Ought-to-be, yet order does not collapse**: If no observer executing boundary creation can be traced, no ought-to-be framework constructed, yet order can spontaneously and stably persist—this system is epistemologically falsified.
2. **[Ontological / Dynamical Level Deadline] No constraints, yet order does not dissipate**: If no rigid constraints are defined inside the ought-to-be framework, yet order has no structural entropy increase or dissipation—this system is physically/ontologically falsified.
3. **[Evolutionary / Developmental Level Deadline] No residuals, yet order can evolve**: If the ought-to-be framework can accomplish generational evolution under constant zero residual norm—this system is evolutionarily falsified.
4. **[Epistemological / Boundary Medium Level Deadline] No boundary creation, yet observer directly accesses actual reality**: If an observer, without passing through boundary creation medium and ought-to-be framework, can directly, fully, and losslessly know and

govern the full picture of physical actual reality—this system is falsified on first principle epistemology.

Chapter 4: Coexistence of Multiple Boundaries and Emergence of Systems

4.1 Physical Encounter of Single and Multiple Boundaries

There is not only one observer drawing one boundary in the world.

When thousands of locals observe independently, draw boundaries independently, and establish respective ought-to-be frameworks, locals inevitably encounter each other physically in finite spatiotemporal space.

4.2 Exclusivity and Hard Collisions

Heaven and earth are finite, and all entities possess exclusivity. Two entities cannot occupy the exact same spot at the exact same moment.

Demands at one place mean reductions at another; expansion at one place means squeezing at another. Exclusivity is the hardest physical reality when multiple boundaries encounter.

4.3 Entanglement and the Birth of Systems

When boundaries of thousands of locals interlock and conflict with each other, entrance and exit of a local affect the entire body:

A gap in one local triggers oscillation in neighboring locals, and oscillation in neighboring locals translates into global violent changes.

This network of mutual entanglement, where physical exclusivity weaves into a web and moving one hair pulls the whole body, is called a “system.”

4.4 Push Orientation and Counteracting Force Collision (Intent Vectors, Inner Product, and Vector Cancellation)

When multiple boundaries coexist, local entities push outward to maintain their ought-to-be frameworks.

Every pushing force has both magnitude and target direction, termed in mathematical physics [Intent Vector].

When two pushing forces meet, same orientation doubles joint force, while opposite orientation cancels each other out; the mathematical projection measuring orientation angle is [Inner Product].

When locals reach thousands, pushing forces of locals point in various directions and counter-strike each other. Cancelled forward forces do not vanish; according to conservation laws, they convert into internal friction heat—this state of local force collision and computational divergence collapse is called “complexity”, termed in physical work [Vector Cancellation and Physical Waste Heat ΔW_{heat}].

Thus, “system”, “complexity”, and full sets of dynamical concepts are not borrowed externally, but naturally grow natively from the three axioms of boundary creation, exclusivity, and multi-boundary encounter.

Chapter 5: End-of-Volume Isomorphism: Continuing the Original Inquiries of Ancient Sages

The boundary creation, ought-to-be, and residual closed loop established by this system is not an isolated armchair conception, but the unfinished inquiry pointed to by scientific philosophy and mathematical logic pioneers of human civilization in pure epistemological dimensions.

I. Immanuel Kant — "Human Mind Legislates Nature" and Subject A Priori Categorical Boundary Creation

Kant explicitly proposed in the *Critique of Pure Reason* that "the human mind legislates for nature," deeply realizing that human cognition is not a passive mirror of the external world, but that the observer subject takes the initiative to delineate boundaries with a priori cognitive categories (the a priori boundary creation operator Π). The "residual-driven second-order evolution" revealed by the third axiom of this system provides Kant's epistemology with a cybernetic dynamical self-healing and model rewriting mechanism.

II. George Spencer-Brown — Laws of Form and Primitive Boundary

George Spencer-Brown resolutely declared as the first axiom in his foundational work of formal logic, *Laws of Form* (1969): "Draw a distinction," mathematically proving that all formal axiomatic systems are born from this first boundary-creating kinetic energy. Axiom 0 (Observer Emergence Law) and Axiom 1 (Generation Axiom) of this system are strictly dual and isomorphic to this classic in formal algebra and topological manifolds.

III. Karl Popper — "Falsifiability" and Residual-Driven Paradigm Rewriting

Karl Popper dispelled positivist inductivism and established the trial-and-error evolutionism driven by counterexamples and residuals in scientific revolutions. The third axiom of this system, "without residuals, order stagnates; with self-healing residuals, order sustains," strictly formalizes Popper's counterexample falsifiability drive into the anti-entropy transition dynamics of differential residual functional norm difference $\Delta = \| X_{\text{real}} - X_{\text{target}} \|$.

> **Epistemological Isomorphism Declaration**: The three pioneers above completed the original inquiries into "boundary creation and self-sustaining order" from the dimensions of epistemology and mathematical logic. Modern scientific and cybernetic isomorphisms (such as Newton, Wheeler, Schrödinger, Shannon, Wiener, Ashby, Qian Xuesen, Wu Xuemou, Cao Longbing, etc.) await completion after formal operators, five-dimensional dual helix, and industrial verdict data are presented in the final volume (Volume V, Chapter 3).

Volume II: Formal Work and Algorithms

— Evolution of the Ought-to-be, Paradigm Breakthrough, and Work Operators

Volume Prelude: Logical Deduction from Three Axioms to Four Work Operators

Volume I *Order* established first principles of order generation, persistence, and evolution from origin. However, the mission of science lies in advancing from axioms to work, from logical deduction to algorithm specifications.

Facing open complex giant systems of factorial complexity $\mathcal{O}(N!)$, without rigorous mathematical operators and state space manifolds, axioms cannot be transformed into actionable cybernetic code.

Deduction relations from Volume I three axioms to Volume II four work operators are as follows:

- **Generation Axiom** (Boundary creation establishing order → collapse without ought-to-be) ⟹ Formally derives **Prior Boundary Creation Operator** Π (dimension-reduction projection of Vol I 1.1 partitioning rights in phase space Ξ);
- **Persistence Axiom** (Rigid constraints → dissipation without constraints) ⟹ Formally derives **Rigid Trajectory Manifold** $\Pi_{\perp}$ (smooth survival channel bounded by Vol I 1.2 hard rules);
- **Evolution Axiom** (Residual feedback → stagnation without residuals) ⟹ Formally derives **Residual Norm Difference** Δ (absolute measure of gap between Vol I 1.2 expectation and actual reality) and **Second-Order Metacognitive Constitutional Amendment Operator** Φ (re-boundary creation work formula jumping out of Vol I 2.3 Gödel deadlock).

This volume aims to express natively grown axioms and concepts of Volume I as formal work operators $\langle \Pi, \Pi_{\perp}, \Delta, \Phi \rangle$, laying hard-core computable algorithm specifications for system and complexity science.

Chapter 1: Evolution of the Ought-to-be World and the Four-Hundred-Year Scientific Impasse

1.1 Physical Essence of the Ought-to-be World

Volume I established: partition boundaries, ought-to-be manifests.

The ought-to-be world is not an entity a priori existing in objective universe, but a local orderly framework constructed by observer through boundary creation in infinite disorderly actual reality background. All expressed, measured, and computed by observer belongs to the ought-to-be world.

Boundary creation action decides “computability” of framework by logic, “calculation” of evolution by rules, and “boundary” of phase space by rigid constraints.

In physical essence, ought-to-be world is negative entropy projection established by observer in local spatiotemporal space to resist spontaneous dissipation and disorder. Without ought-to-be framework, order cannot attach, and everything inside boundary collapses.

Brilliance and Physical Impasse of Four-Hundred-Year Reductionist Paradigm: Over past 400 years, humanity attempted to construct ought-to-be order using reductionism, forcibly assuming systems possess microscopic single-point independence, static lead times, and linear additivity. In simple systems, this disassembly is self-consistent; however, when facing open systems with interlocking nodes and dynamic games, static rules of reductionism instantly fail. Systems emerge sea of conflicts, and phase space combination possibilities explode factorially. Humanity must answer: what are necessary orthogonal bases that phase space must possess geometrically to completely and losslessly describe and govern ought-to-be world in physical spatiotemporal and causal networks? This derives physical five-dimensional minimal completeness in mathematics and physics.

1.2 Physical Five-Dimensional Minimal Completeness Theorem (Theorem 1.1)

According to five orthogonal bases naturally grown from boundary creation and phase space in Vol I 1.2, observer's boundary creation action necessarily and uniquely derives ought-to-be topological manifold $\mathcal{M}_{\text{model}} = \langle \mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X \rangle$.

- **Necessity (Missing one dimension collapses structure)**: Stripping N (Nodes) → no objects; stripping T (Topology) → no order; stripping C (Constraints) → no boundaries; stripping X (States) → no time; stripping ΔX (Residuals) → no feedback.
- **Sufficiency (Self-consistent closed loop)**: When five orthogonal bases are fully possessed and constraint compatibility satisfies non-empty feasible domain ($\text{Sol}(\mathcal{C}) \neq \emptyset$), ought-to-be system extreme closed loop is necessary and sufficient. **[Plain Explanation]**: "$\text{Sol}(\mathcal{C}) \neq \emptyset$" in plain words means **"this tough problem has at least one legal feasible solution, rather than mutually contradictory unresolvable deadlock"** (e.g., cannot demand a horse to run fast while eating no grass). Only when constraint conditions are mutually compatible and feasible solution set is non-empty can system truly run in physics.

Conclusion: Necessity + Sufficiency ⇒ $\mathcal{M}_{\text{model}}$ is minimal complete phase space basis for ought-to-be system.

> **[Statement on "Rigid-Flexible Nested" Essence of Five-Dimensional Manifold]**: Physical five-dimensional orthogonal basis $\mathcal{M}_{\text{model}} = \langle \mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X \rangle$ is not a homogeneous static set, but presents **"rigid geometric base manifold + deformable dynamic tangent fiber" double-layer nesting** in topology:
>
> 1. **First three dimensions $\langle \mathcal{N}, \mathcal{T}, \mathcal{C} \rangle$ are rigid structural base**: Defining discrete node occupation, causal topology (DAG), and exclusive conservation constraint clusters, establishing terrain and hard rules of phase space, providing non-backtracking causal pipelines for polynomial closed-loop computability;
> 2. **Latter two dimensions $\langle X, \Delta X \rangle$ are continuous variable flow fibers**: X describes continuous stretching and configuration deformation of system potential energy distribution, $X_{\text{plan}} = X \oplus \Delta X$ describes state transitions and energy flow along causal topology pointers; when perception residual breaks critical threshold, ΔX mutates into phase transition breakdown operator, reversely driving generational reconstruction (second-order amendment) of first three structural bases. **Rigid base guarantees computation (persistence), flexible fiber guarantees adaptation and leap (evolution).**

Chapter 2: Paradigm Breakthrough: Hundred-Billion-Scale Empirical Validation and Computable Work

2.1 Ultimate Question from the Physical Battlefield

On hundred-billion-scale discrete manufacturing physical battlefields, characteristics of open complex giant systems are fully displayed:

Tens of thousands of physical execution units, massive heterogeneous multi-level kitting constraints, dynamically fluctuating time limits, and sudden anomalies. Combination possibilities of phase space explode factorially ($\mathcal{O}(N!)$) with node permutations.

Facing factorial complexity, four-hundred-year reductionism static rules completely fail. The ultimate question: how to achieve millisecond self-healing closed-loop and "human-out-of-the-loop" autonomous decisions under physical laws?

2.2 Paradigm Breakthrough: Self-Consistency as Necessary and Sufficient Guarantee for Closed-Loop Computability

This system does not originate from applying abstract academic concepts, but from real physical work. Facing factorial complexity $\mathcal{O}(N!)$ of physical battlefields, without burdens of post-hoc academic jargon and theoretical frameworks. Starting from plain origin along shortest logical path, through algebraic pruning of prior boundary operator Π, logic self-consistency of system physically and sufficiently guarantees "intra-boundary closed-loop computability and extreme uniqueness" of open complex giant systems.

Core of this breakthrough lies in completing system science **paradigm leap from "descriptive philosophy" to "work science"**:

1. **Rejecting empty qualitative descriptions, establishing rigorous operators and physical constraints**. We are not satisfied with exclaiming "the whole is greater than sum of parts", but through prior boundary creation operator Π, forcibly strip uncomputable and uncontrollable redundant degrees of freedom in phase space, performing algebraic pruning on factorial complexity $\mathcal{O}(N!)$.
2. **Rejecting open-loop static predictions, establishing dynamic feedback closed-loop based on residuals**. We established residual norm difference Δ **as sole driving force of evolution**. Systems no longer obsess over maintaining zero-fluctuation static steady states, but convert external impacts and passively triggered residuals into evolutionary energy driving system metacognitive constitutional amendment operator Φ to rewrite paradigms.

Based on this engineering empirical proof, we proved "intra-boundary closed-loop computability and extreme uniqueness" of open complex giant systems under boundary creation conditions. This self-consistent path laid solid physical and engineering bedrock for subsequent work operators and algorithm specifications.

Chapter 3: Formal Work Operators and Topological Manifolds

3.1 Prior Boundary Creation Operator Π and Phase Space Truncation

Vol I Generation Axiom established: partition boundaries, ought-to-be manifests; without ought-to-be framework, everything collapses.

In formal work mathematics, absolute starting action of observer boundary creation is formally defined as prior boundary creation operator Π.

Infinite, uncountable unpartitioned actual reality background is expressed as infinite-dimensional phase space Ξ. Prior boundary operator Π is a dimension-reduction projection operator acting on phase space Ξ, whose physical work manifests as cropping infinite phase space into finite ought-to-be state space Ω containing physical five dimensions:

$$\Pi: \Xi \to \Omega(\mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X)$$

Where $\dim(\Omega) \ll \dim(\Xi)$. Boundary operator Π possesses idempotency ($\Pi^2 = \Pi$) and redundant degree of freedom stripping properties, forcibly locking computable and calculable orthogonal bases inside boundary.

Boundary creation operator Π possesses three core algebraic and physical properties:

1. **Idempotency (Framework Confirmation)**: $\Pi^2 = \Pi$ Once boundary creation action is completed, ought-to-be framework boundary is determined. Observer executing boundary creation again inside same ought-to-be framework does not change established boundary topology.
2. **Redundant Degree of Freedom Stripping**: Operator Π forcibly strips uncomputable, uncontrollable redundant degrees of freedom in phase space, retaining only computable, calculable orthogonal bases inside boundary. Without dimension-reduction projection work of boundary operator Π, ought-to-be framework cannot attach, and everything inside boundary collapses.
3. **Three-Domain Work Manifestation**:
    - In perception domain: Operator Π manifests as filter mapping, converging infinite physical noise into high-fidelity perception residuals;
    - In computation domain: Operator Π manifests as topological pruning, forcibly locking finite evolution trajectories in phase space;
    - In action domain: Operator Π manifests as boundary constraints, forcibly converging invalid degrees of freedom of microscopic nodes.

3.2 Rigid Trajectory Manifold $\Pi_\perp$ and Persistence Work

Vol I Persistence Axiom established: without rigid constraints and rules, order inside boundary inevitably dissipates.

In state space Ω, if rigid constraints are not imposed on evolution trajectories of state vectors, system states spontaneously diffuse along disorderly directions, manifesting in physical domain as spontaneous entropy increase ($d_i S > 0$), and effective energy inevitably dissipates in internal friction.

Prior boundary creation operator Π further derives rigid trajectory manifold $\Pi_\perp$ inside state space Ω:

$$\Pi_\perp \subset \Omega$$

Manifold $\Pi_\perp$ is a low-dimensional closed manifold bounded by rules and physical constraints in state space. System state vector X during evolution is forcibly confined onto manifold $\Pi_\perp$:

$$X \in \Pi_\perp, \quad \forall t \geq 0$$

When system is impacted externally and deviates from rigid trajectory manifold $\Pi_\perp$, normal manifold constraint forces perform corrective work. Its physical essence: stripping non-orthogonal degrees of freedom normally, forcibly counteracting spontaneous entropy increase, achieving anti-entropy persistence of system.

In physical work mechanism, intent vectors $\boldsymbol{v}_i = \frac{\mathrm{d}x_i}{\mathrm{d}t}$ of nodes along state transition directions produce physical phase interference, resulting in joint vector magnitude strictly smaller than sum of independent magnitudes (vector triangle inequality): $\left\|\sum_{i=1}^{N} \boldsymbol{v}_i\right\| < \sum_{i=1}^{N} \| \boldsymbol{v}_i\|$

Difference between the two is exact physical work cancellation of force collision in phase space described in Vol I, completely converting into internal friction waste heat of system:

$$\Delta W_{\text{heat}} = \sum_{i=1}^{N} \| \boldsymbol{v}_i \| - \left\| \sum_{i=1}^{N} \boldsymbol{v}_i \right\| \geq 0$$

Work essence of rigid manifold $\Pi_\perp$ is precisely eliminating vector cancellation by stripping non-orthogonal degrees of freedom normally ($\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle \geq 0 \Longrightarrow \Delta W_{\text{heat}} \to 0$), achieving orthogonal decoupling and first-order self-healing inside system.

Variational Extremum and Work Internal Friction Minimization Theorem Established:

1. **Variational Extremum and Work Internal Friction Minimization Uniqueness Theorem ($\exists! S_k^*$)**: Constraint manifold $\mathcal{C} = \mathcal{C}_{\text{hard}} \cup \mathcal{C}_{\text{soft}}$ rigidly encodes entity spatiotemporal exclusivity laws, target commitment boundaries (excluding trivial idle zero-work $\boldsymbol{v}_i = \boldsymbol{0}$), and vector cancellation elimination conditions ($\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle \geq 0$). For any feasible trajectory $S \in \mathrm{Sol}(\mathcal{C})$ walking through both hard and soft constraints, total system loss decomposes orthogonally as $W_{\text{loss}}^{\text{total}}(S) = W_{\text{intrinsic}}(\mathcal{C}_{\text{hard}}) + \mathcal{W}_{\text{heat}}(S)$, where $W_{\text{intrinsic}}(\mathcal{C}_{\text{hard}})$ is the irreducible physical floor dictated by fixed hard constraints and $\mathcal{W}_{\text{heat}}(S) \geq 0$ is non-negative friction waste heat established by the vector triangle inequality. Algorithm $\mathcal{A}$ deduces forward along the causal DAG and prunes conflicting branches normally; whenever S_k^* traverses inside the boundary, cumulative waste heat vanishes ($\mathcal{W}_{\text{heat}}(S_k^*) = 0$). By contradiction, any feasible walked-through trajectory S' achieving strictly lower total loss $W_{\text{loss}}^{\text{total}}(S') < W_{\text{loss}}^{\text{total}}(S_k^*) = W_{\text{intrinsic}}(\mathcal{C}_{\text{hard}})$ would require $\mathcal{W}_{\text{heat}}(S') < 0$, violating the vector triangle inequality; combining rigid spatiotemporal exclusivity with a strict lexicographic tie-breaking order $(t_{\text{avail}}, T_{\text{commit}}, -\mathcal{P}, \text{ID})$ for simultaneous equal-urgency elements eliminates symmetric permutation ties, guaranteeing convergence to a unique in-boundary global optimal steady state ($\exists! S_k^* \in \mathrm{Sol}(\mathcal{C})$).
2. **Causal Finiteness Lemma 1.1 and Two-Stage Non-Backtracking Computability ($K < +\infty$)**: On a bounded-in-degree causal DAG where shared intermediate nodes merge rather than duplicating as trees ($|\mathcal{E}| = \mathcal{O}(N)$), accumulable extensive flows satisfy forward temporal persistence (once ready at t_{avail}, they remain persistently valid across $t' \in [t_{\text{avail}}, T_{\text{commit}}]$); hence Stage 1 linear kitting propagation ($\mathcal{O}(N)$) and Stage 2 local residual-window sorting of perishable spatiotemporally exclusive intensive capacities ($\mathcal{O}(N\log N)$) are unidirectionally decoupled with zero cross-stage combinatorial backtracking (single-step atomic evolution operator $P_k \sim \mathcal{O}(N\log N)$). Coupled with discrete physical clock step $\Delta t_{\text{clock}} > 0$ and instability relaxation window $\tau_{\text{phy}} < +\infty$ truncating causal depth to finite steps ($K \leq \frac{\tau_{\text{phy}}}{\Delta t_{\text{clock}}} < +\infty$), this strictly guarantees closed-loop computation time $T(N) = \mathcal{O}(K \cdot N\log N) \in \boldsymbol{P}$ and $T(N) \leq \tau_{\text{phy}} < +\infty$. Because this derivation relies solely on three minimal physical

invariants—bounded-DAG multi-prerequisite kitting gates, extensive-vs.-intensive resource dichotomy, and finite discrete-clock truncation—it holds universally across all Open Complex Giant Systems via structural isomorphism.

3.3 Residual Norm Difference Δ and Second-Order Metacognitive Constitutional Amendment Operator Φ

Vol I Evolution Axiom established: without residual feedback, generational evolution stagnates.

Residual is formally expressed as state norm difference Δ in work mathematics:

$$\Delta = \| \Omega_t - \Omega_{t-1} \|$$

Where Ω_t represents actual reality observation mapping of current time step, Ω_{t-1} represents ought-to-be expectation of previous time step. Regardless of whether residual Δ is zero, it is monitored in real time by observer.

When external environment paradigm undergoes drastic phase transition, causing residual Δ to break preset safe critical evolution threshold (formally denoted θ_{trigger}):

$$\Delta > \theta_{\text{trigger}}$$

Residual is forcibly triggered, and system leaps from first-order ought-to-be calculus to activate second-order metacognitive constitutional amendment operator Φ.

Second-Order Metacognitive Constitutional Amendment Operator Φ does not make local parameter adjustments inside current ought-to-be framework, but transcends current formal axiomatic constraints to rewrite and reconstruct orthogonal bases of prior boundary operator Π and rigid manifold $\Pi_\perp$:

$$\Phi: \Pi_k \to \Pi_{k+1}$$

Through second-order metacognitive constitutional amendment work of operator Φ, rigidity and dogma of old frameworks are broken, promoting system paradigm leap and generational evolution. (For concrete operational mechanisms of rewriting physical five-dimensional matrix values under residual drive, see Appendix B.1.)

Chapter 4: Embodiment of Operators: Five-Dimensional Topological Mind and Neural Mapping

4.1 Embodied Mapping of Formal Operators in Carbon-Based Brains and Control Systems

Formal work operators established in previous three chapters are not floating abstract symbols; in biological carbon-based neural systems and high-order control systems, they completely embody into "conscience-driven five-dimensional mind topology" (conscience, hypersensitivity, affect/intuition, fluid intelligence, metacognition).

Self-organizing open systems, to maintain survival boundaries, evolved second-order self-referential awareness. There exists a 1:1 absolute topological mapping between work operators and biological neural networks:

Formal Work Operator / Physical Mechanism	Embodied Mind Dimension	Neural Network & Brain Region Topology Mapping
Conscience Compact Support Operator E_{supp}	Conscience	Top-level Bayesian Prior & Default Mode Network (DMN)
Precision-Weighted Radar	Hypersensitivity	Amygdala-Locus Coeruleus (AMY-LC-NE) Arousal System
Heuristic Reconstruction Cache	Affect / Intuition	Salience Network (SN)
Phase Space Matrix Solution	Fluid Intelligence	Frontoparietal Control Network (FPN)
Second-Order Metacognitive Constitutional Amendment Operator Φ	Metacognition	Frontopolar Cortex (BA10 Area)

4.2 Operator E_{supp} (Conscience/Top-Level Prior) and Goodhart Collapse Truncation

In full-brain hierarchical predictive architectures, conscience is represented as highest top-level Bayesian prior, with hardware foundation being brain's Default Mode Network (DMN).

When optimization pressure releases toward infinity, deviation Δ occurs between proxy indicators and true intent. If probability density function $p(\Delta)$ of deviation Δ exhibits heavy-tailed characteristics, according to Goodhart's inequality, limit optimization causes expected utility of true goal to diverge to negative infinity: **[Plain Explanation]**: As common saying goes, "when a measure becomes a target, it ceases to be a good measure" (Goodhart's Law). For example: if programmers are evaluated solely on "lines of code typed", they will crazily copy-paste useless redundant code. Mathematically, when infinite optimization pressure is applied to a local indicator, people crazily exploit indicator loopholes for arbitrage, causing proxy indicators to severely deviate from true targets, ultimately triggering overall system collapse. This is direct physical meaning of Goodhart's inequality.

$$\lim_{\text{optimization}\to\infty} E[r^*] = -\infty$$

In mathematical control layers, conscience is formally defined as **compact support operator** E_{supp} acting on residual probability density $p(\Delta)$.

Operator E_{supp}, at physical pain trigger critical points, forcibly restricts support set of residuals inside a compact interval $[-\theta_{\text{dead}}, \theta_{\text{dead}}]$, truncating heavy-tailed distributions into light-tailed sub-Gaussian distributions:

$$E_{\text{supp}} \cdot p(\Delta) \in \text{Sub-Gaussian}$$

This mathematically guarantees analytical convergence of integrals, forcibly severing pathways of indicator arbitrage and system hacker attacks from mechanism level, maintaining top-level alignment of ought-to-be frameworks.

On system meta-architecture, conscience operator E_{supp} does not belong to local rules of first-order ought-to-be calculus, but is a [Second-Order Physical Meta-Governing Principle] hanging above holographic metacognition and evolutionary algorithm $\mathcal{A}$. Physical five dimensions complete orthogonal decoupling and eliminate vector cancellation inside system; while conscience operator E_{supp} anchors ultimate gravity above system, forcibly truncating residual heavy-tailed distributions $E_{\text{supp}} \cdot p(\Delta) \in$ Sub-Gaussian, ensuring first-order self-healing system never crosses generational

survival red lines when leaping toward limit optimization. (For formal mathematical proof of Goodhart inequality divergence and E_{supp} compact support truncation equations, see Appendix A.2.)

4.3 Hypersensitivity Precision-Weighted Radar and Affective Heuristic Cache Work

In perception and feedback loops, hypersensitivity and affect constitute neural twin wings of system capturing and responding to residuals in real time:

1. **Hypersensitivity Precision-Weighted Radar (Perception)**: Hypersensitivity in computation layer is essentially precision weighting (so-called precision weighting means brain does not treat all perceptual signals equally, but amplifies key signals according to survival importance, amplifying microscopic gaps into actionable perception residuals) mechanism applied by full-brain system on perception channels. Its hardware foundation is amygdala-locus coeruleus-norepinephrine system (AMY-LC-NE). Hypersensitivity radar turns up synaptic transmission gain, amplifying microscopic physical gaps into holographic perception residuals $\Delta(t)$, injecting initial energy required for driving system.
2. **Affective Heuristic Cache (Feedback)**: Affect in computation layer manifests as System 1 (so-called System 1 refers to fast, automatic, low-energy intuition response system in brain; affective heuristic cache refers to System 1 compressing past work experience into ultra-low energy reflex responses) emotion and heuristic caching, whose core hub is Salience Network (SN) composed of anterior insula (AIC) and anterior cingulate cortex (ACC). It compresses past dimension-reduction work experience into ultra-low energy reflex responses, forcing fluid intelligence to execute limit optimization in phase space.

4.4 Fluid Intelligence Matrix Solution and Second-Order Metacognitive Constitutional Amendment Operator Φ (BA10 Evolution)

First-order ought-to-be calculus facing environment of factorial complexity $\mathcal{O}(N!)$ inevitably falls into algorithm halting and logical deadlocks. Systems must achieve evolutionary leaps transcending deadlocks through fluid intelligence and second-order operator Φ:

1. **Fluid Intelligence Matrix Solution**: Its hardware foundation is Frontoparietal Control Network (FPN). When residual Δ appears and cannot be resolved by affective heuristic cache, fluid intelligence is forcibly activated, solving corrective work of state vector X and rigid trajectory manifold $\Pi_{\perp}$ in working memory.
2. **Second-Order Metacognitive Constitutional Amendment Operator Φ and Frontopolar Cortex (BA10 Area) Evolution**: Human brain's frontopolar cortex (BA10 area) expanded 4.7 times in relative volume compared to chimpanzees, evolving second-order metacognitive constitutional amendment operator Φ transcending first-order algorithm deadlocks. When system faces logical deadlocks and singularities, second-order operator Φ transcends current formal axiomatic constraints, re-drawing prior operator Π to complete paradigm rewriting of axiomatic bases:

$$\Phi: \Pi_k \to \Pi_{k+1}$$

This proves second-order metacognitive constitutional amendment operator Φ is precisely the specific evolutionary operator for carbon-based biological brains to cross Gödel incompleteness boundaries and Turing halting boundaries.

Volume III: Governing Principles and Dimension-Reduced Engineering Code

— Dimension-Reduced Projection of System Engineering

Volume Prelude: Causal Chains and Mathematical Mechanisms of Four Operators Projecting into Eight Treatises

Volume II established four prior work operators (Prior Boundary Creation Operator Π, Rigid Trajectory Manifold Operator $\Pi_{\perp}$, Residual Norm Difference Operator Δ, and Second-Order Metacognitive Constitutional Amendment Operator Φ) and algorithm specifications. This volume aims to deduce rigorous mathematical mechanisms of four work operators projecting down into phase space of system engineering and entity governance.

[Mathematical Medium Mechanism of Dimension-Reduction Projection and "Eight Treatises" Academic Naming Declaration]:

1. **"Eight Treatises" Academic Naming Declaration**: Use of "Eight Treatises" (Teleology, Ontology, Scheme, Capability, Mechanism, Path, Dynamics, Evolution) in this system strictly corresponds to **"eight rigorous formal system treatises and causal deductions"** in cybernetic and scientific discourse contexts (i.e., Treatise on Establishment of Teleology, Treatise on Compensation of Ontology, Treatise on Equivalence of Scheme, Treatise on Singularity of Capability, Treatise on Decoupling of Mechanism, Treatise on Foundation of Path, Treatise on Contract of Dynamics, Treatise on Amendment of Evolution). Clarification: "Treatise" here is a verb meaning "treatise/deduction", by no means static discipline or school labels in metaphysical scholastic philosophy (such as traditional philosophy "Teleology" or "Evolutionism"). It is a formal cybernetic remedy for eight causal gates open complex giant systems must pass through across full life cycles of generation, persistence, and evolution, starting from physical first principles.
2. **Mathematical Dimension-Reduction Medium**: Mapping medium of four atomic work operators projecting down into eight formal treatises of system engineering is precisely **physical five-dimensional manifold** $S = \mathcal{M}_{\text{model}} \otimes \mathcal{A}$ established in Volume II (continuous/discrete hybrid manifold composed of Nodes $\mathcal{N}$, Topology $\mathcal{T}$, Constraint Clusters $\mathcal{C}$, State Vector X, and State Transitions with causal residuals ΔX). "Eight Treatises" are by no means empirical management classifications, but strict mathematical projections of four work operators under topological geometric constraints of physical five-dimensional manifold along orthogonal observation planes and control boundaries.

An objective physical reality must be observed: spontaneous evolution direction of any open complex giant system is order decay and entropy increase ($d_i S > 0$). In entity phase space of factorial complexity $\mathcal{O}(N!)$, linear management mechanisms based on local experience suffer computational bottlenecks and dimension deficiencies.

Four operators projecting down into system engineering phase space sequentially evolve into eight-treatise causal closed loop of system governance (each treatise being deadlock remedy for previous one):

1. **Teleology** (Boundary creation establishing all-domain potential energy field and holographic negative entropy attractor) → but global solving encounters $\mathcal{O}(N!)$ illusion and human brain channel limits →

2. **Ontology** (Human-out-of-the-loop, silicon computation compensating Metasynthetic Engineering / Hall for Workshop) → but business and IT division fragment causing information distortion and loss (cross-interface Shannon entropy increase) →
3. **Scheme** (Business ontology, system carrier, algorithm model trinity fusion capability) → but cross-domain capability blocked →
4. **Capability** (Fusion capability and single-brain origin, stripping degrees of freedom normally) → but multi-head consultation deadlock and committee decision paralysis →
5. **Mechanism** (Nodes freely inputting demands, free input with Metasynthetic Engineering logical autocratic convergence, decision automated write-back) → but lacking closed-loop work route →
6. **Path** (Establishing human-machine dual ought-to-be carriers, following path of "unobservable means uncontrollable, building backward from controlled end") → but non-bottleneck crude work and phase angle interference →
7. **Dynamics** (Eliminating vector cancellation and Dual-Chief System supreme contract) → but first-order rules encounter environment phase transition falling into Gödel deadlock and Turing halting →
8. **Evolution** (Activating active state second-order metacognitive amendment operator Φ generational leap).

Governing principles deduced in this volume aim to demarcate objective boundaries for intelligent subjects executing dimension-reduced governance on complex physical reality: algorithms define order, computation executes compensation, and conscience anchors gravity.

Chapter 1: Teleology: Three Physical Pillars of Holographic Anti-Entropy

1.1 Extremal Attractor and Holographic Negative Entropy

Spontaneous evolution direction of open complex giant systems naturally points to entropy increase and dissipation. Teleology is not subjective speculation, but **holographic negative entropy extremal attractor** established by system in phase space. Just as light rays automatically travel along least-time paths and water droplets automatically converge to lowest surface tension configurations, establishing all-domain potential energy fields in system essentially marks "geodesics" with minimal internal friction for all microscopic nodes in phase space. Without all-domain attractors, systems inevitably fall into disorderly diffusion and spontaneous collapse.

1.2 Vector Cancellation Law and Physical Waste Heat

In systems lacking all-domain potential energy field governing, microscopic nodes perform work independently under local interest drives. Intent vectors of nodes present mutual angles or counter-strike in phase space, causing macroscopic joint work magnitude to be strictly smaller than sum of independent magnitudes. Work energy cancelled out by vector cancellation completely converts into internal friction waste heat of system in physics:

$$\Delta W_{\text{heat}} = \sum_{i=1}^{N} \| \boldsymbol{v}_i \| - \left\| \sum_{i=1}^{N} \boldsymbol{v}_i \right\| \geq 0$$

This is qualitative physical origin of internal friction heat generation in multi-node systems.

1.3 Extremal Closed-Loop of Governing Work

Prior boundary creation operator Π establishes global attractors, whose qualitative physical essence lies in eliminating phase angle interference and vector cancellation waste heat ($\Delta W_{\text{heat}} \to 0$) by absorbing non-orthogonal components, achieving complete convergence of system effective work toward holographic goals.

Teleology Governing Principle Established: Global potential energy fields and holographic attractors are ultimate anchors for systems resisting spontaneous dissipation. Qualitative physical essence lies in establishing orthogonal bases through prior boundary creation, eliminating vector counter-strike waste heat of microscopic nodes, achieving optimal closed loop of system all-domain work conversion rate.

Chapter 2: Ontology: Physical Essence from Single-Point Reductionism to Human-Machine Symbiosis

2.1 Applicable Boundaries of Simple Systems and Reductionism

Vol I definitions noted: actual reality silently occurs; partition boundaries, ought-to-be manifests.

In initial stages of system evolution, when inside boundary contains very few independent nodes, system is in simple phase state.

In simple phase states, interactions between elements exhibit linear additivity. Reductionism paradigm built by humanity over past 400 years maintains high self-consistency within this boundary: by disassembling system into single-point microscopic units and solving static equations, behavior of entire system can be accurately predicted and controlled.

Qian Xuesen System Taxonomy and Degeneration Theorem Established: Qian Xuesen categorized systems into simple systems, simple giant systems, complex giant systems, and open complex giant systems (OCGS). Among them, simple systems and simple giant systems possess linear additivity and homogeneous features, being degeneration limits of open complex giant systems in physical phase space. Within simple system boundaries, solving static equations by disassembling microscopic single points maintains high self-consistency. This system completely incorporates Qian's system taxonomy, proving linear additive systems are degenerate special cases under complex evolution, achieving all-domain physical coverage of "all order from simple systems to open complex giant systems".

2.2 Node Coupling and Emergence of Factorial Complexity $\mathcal{O}(N!)$

When node scale expands and non-linear coupling games exist, system undergoes spontaneous phase transition, emerging as open complex giant system of factorial complexity Complexity $\sim \mathcal{O}(N!)$.

When node quantity N grows, possibilities of interactions and games between nodes explode factorially $\mathcal{O}(N!)$:

$$\text{Complexity} \sim \mathcal{O}(N!)$$

Facing factorial complexity, **disassembly logic of reductionism completely bankrupts**: simple superposition of microscopic single points cannot derive global emergence, static lead time assumptions are washed away by non-linear fluctuations, and open-loop human meetings and empirical decisions diverge and collapse in physics.

2.3 Human-Machine Symbiosis and Computational Compensation

When phase space complexity reaches $\mathcal{O}(N!)$, biological carbon-based brains are limited by neuron conduction speeds and thermodynamic energy budgets, physically unable to solve global optimal trajectories.

Ontology Governing Principle Established: Physical essence of open complex giant system governance lies in abandoning blind intervention of carbon-based single points, moving toward "human-out-of-the-loop, human-machine synergy".

Intelligent subjects must utilize extremely fast concurrent computation of silicon systems to execute algebraic pruning of prior boundary operator Π in computation space, compensating physical limits of carbon-based brain power with silicon computation power, achieving self-healing closed loops of complex entities on physical battlefields.

Chapter 3: Scheme: Five-Dimensional Dual-Helix and Dimension-Reduced Engineering Scheme

3.1 Formal Confirmation of Physical Five-Dimensional Basis

Vol I Generation Axiom established: without ought-to-be framework, everything collapses. Governance schemes of any open complex giant system physically and sufficiently build upon five orthogonal axiomatic primitive elements:

1. **Nodes ($\mathcal{N}$)**: Discrete or continuous particle elements independently carrying states in phase space;
2. **Topology ($\mathcal{T}$)**: Directed coupling network matrices A of interactions and energy/information transmission between particles;
3. **Constraints ($\mathcal{C}$)**: Rigid extrema and geometric cropping planes C limiting feasible evolution orbital domains of phase space;
4. **State Vectors (X)**: Phase potential energy coordinates and field distribution vectors of system under instantaneous time slices. Internally containing bi-directionally aligned physical potential differences: Target Tension Vector X_{target} (attractor pull) and State Potential Vector X_{state} (base potential energy);
5. **State Transitions (ΔX)**: State trajectory evolution operators governed by causality and driven by dynamic work.

Without physical confirmation of physical five dimensions, all system engineering schemes are water without source, trees without roots. (For mathematical definitions of physical five dimensions as phase space topological invariants and isomorphic mappings with Prof. Longbing Cao's Non-IID theory, see Appendix A.3 and Appendix C.1.)

3.2 Solution Scheme and Dimension-Reduced Work of the Five-Dimensional Dual-Helix

On physical battlefields of factorial complexity $\mathcal{O}(N!)$, dimension-reduced governance schemes manifest as interwoven work of five-dimensional dual helix:

- **Holographic Data Model Helix**: Accurately mapping [Nodes, Topology, Constraints] into rigid manifold $\Pi_{\perp}$, loading [State Vector X] and [Historical Causality ΔX_{hist}];

- **Dynamic Evolution Algorithm Helix**: Evolution algorithm $\mathcal{A}$ solves optimal control vector u^* and current [State Transitions with causal pointers and Residuals ΔX], generating [Next Moment New State X_{plan}];

Five-dimensional dual helix solution equations are expressed as:

- Model input before calculation: Defining holographic data model at time t as state tuple $M = \{\Pi_{\perp}, X_{\text{target}}, X_{\text{state}}, \Delta X_{\text{hist}}\}$.
- Solving output during calculation: $(u^*, \Delta X) = \mathcal{A}(M)$
- Recurring new state after calculation: $X_{\text{plan}} = X \oplus \Delta X$

[Physical Reality Declaration of Causal Anchoring and State Transitions with Causal Pointers ΔX]: Plus sign "+" in mathematical formulas is an abstract generalization; in real physical and dynamic networks, causal carrier and topological mapping connecting X to X_{plan} is called **Causal Anchoring**:

1. **States cannot teleport, must have causal carriers**: Microscopic entity configuration X at time t must, through concrete causal actions and physical work, evolve into new state configuration X_{plan} at time $t + 1$ along causal topological directed edges (DAG Edges);
2. **Physical essence of causal anchoring**: In phase space, accurately anchoring microscopic state nodes of X and target demand sites of X_{plan} with rigid causal pointers (*which target does current element perform work for? which evolution operator occupies current site?*);
3. **Forward causal generation equation**:

$$X_{\text{plan}} = X + \text{CausalAnchor}(X)$$

Without causal anchoring, state transitions with causal pointers ΔX are merely soulless statistical numbers; with causal anchoring, ΔX becomes a topological work flow driven by causal pointers.

Scheme Governing Principle Established: Locking ought-to-be boundaries through [Nodes, Topology, Constraint Clusters], solving reverse work through [State Vectors, State Transitions]. Core of scheme lies in establishing **trinity fusion capability of "business ontology, system carrier, and evolution model"**. This scheme is by no means a microscopic local supply chain tool, but a dimension-reduced engineering scheme of open complex giant systems fusing physical business reality, software system carriers, and control algorithm models, forcibly eliminating mismatch angles between intent and laws, achieving dynamic computable closed loop of system engineering schemes.

3.3 Parallel Universe Simulation and Absorption Work of the Five-Dimensional Basis

Before physical entities make real responses, five-dimensional dual helix scheme expands "parallel universe" concurrent solutions in computation space.

Algorithm helix, inside solution space determined by [Topology] and [Constraint Clusters], conducts N sets of concurrent simulations on future [State Transitions] of [State Vectors]:

$$X'_k = \mathcal{F}(X, u_k), \quad k = 1,2,\dots,N$$

Scheme automatically eliminates divergent transitions violating [Constraint Clusters], retaining only optimal dimension-reduced trajectories evolving along rigid manifold $\Pi_{\perp}$, and writing control vectors $u(t)$ back to microscopic [Nodes].

Five-Dimensional Scheme Governing Principle Established: Solving five-dimensional transitions in computation space, harvesting hard-core anti-entropy in physical space.

Model Building Human-Machine Synergy Theorem Established: Not only scheme execution relies on silicon computation compensation, but starting point of building high-dimensional five-dimensional data models and evolution algorithms itself physically and sufficiently relies on human-machine synergy. Carbon-based brain power is limited by energy consumption and calculation bandwidth, unable to solve high-dimensional orthogonal bases independently; pure silicon computation power is limited by Gödel's incompleteness theorem, lacking second-order metacognitive constitutional amendment operator Φ to set prior boundaries. Only "human second-order metacognitive constitutional amendment Φ_{Human} establishing prior boundaries + silicon ultra-fast computation $\mathcal{A}_{\text{Silicon}}$ solving topological constraints" can build lossless five-dimensional dual helix models. "Human-out-of-the-loop, human-machine synergy" is necessary and sufficient physical condition for model building and algorithm evolution.

Chapter 4: Capability: Rigid Trajectory Manifold $\Pi_{\perp}$ Normal Stripping of Degrees of Freedom and Anti-Entropy Carrying Capacity

4.1 Business Understanding Capacity: Orthogonal Abstraction Based on Data Models

Vol I Capability and Second Law (Persistence Axiom) noted: without rigid constraints and rules, order inside boundary dissipates. Capability is physical carrier of system landing schemes.

Primary capability in solving complex entity systems lies in business understanding capability based on data models.

Facing sea of heterogeneous physical resources, execution nodes, and temporal constraints on physical battlefields, control systems must possess capability to abstract physical reality into five-dimensional orthogonal bases. Lacking business understanding capability, data models will confuse constraint conditions, producing non-orthogonal coupling.

Essence of business abstraction capability lies in accurately identifying effective control variables inside boundary through prior boundary operator Π, stripping background physical noise, providing lossless data bases for subsequent control work.

4.2 Architectural Synergy Capacity: Normal Stripping of Redundant Degrees of Freedom by Rigid Manifold $\Pi_{\perp}$

Open complex giant systems contain sea of cross-subsystems. Second core capability lies in cross-subsystem architectural synergy capability.

Disorderly diffusion of microscopic nodes physically manifests as non-orthogonal redundant degrees of freedom $X_{\perp}$ on normal directions. Control system applies normal constraint forces through rigid trajectory manifold $\Pi_{\perp}$, forcibly stripping redundant degrees of freedom:

$$X_{\perp} = X - \Pi_{\perp} \cdot X \to 0$$

Capability Governing Principle Established: Physical essence of architectural synergy capability lies in forcibly eliminating mutual friction and internal consumption between cross-subsystems. When normal degrees of freedom are forcibly stripped to zero, resistance between subsystems is completely flattened, effective work no longer turns into internal dissipation waste heat, and overall system

capability achieves leap from single-point scatter to global orthogonal synergy. (See Appendix A.2 for proof of Goodhart inequality and E_{supp} compact support truncation.)

4.3 Fused Work Capacity and Single-Brain Origin Theorem

Third core capability lies in anti-entropy carrying capability seamlessly fusing "business ontology, system carrier, and algorithm model" trinity.

In biology and cognitive neuroscience, neural networks of different human brains cannot perform direct signal mapping and synergy across skulls. Therefore, high-order paradigm reconstruction and fusion capability **physically and sufficiently expands with "single brain (Chief Designer / Commander)" as origin and hub**.

Control systems do not rely on fragile scattered node negotiations, but establish orthogonal bases with single brain as cognitive hub, absorbing external impacts into network topology through distributed structure topology matrix A:

$$\text{Capacity} \sim D(A)$$

Where $D(A)$ is redundancy absorption degree of topology matrix.

Capability and Single-Brain Origin Governing Principle Established: Capability of strongly robust systems lies in completing high-order fused orthogonal abstraction with single-brain cognition as starting point, instantly flattening and absorbing local anomalies. Systems through single-brain governed microscopic nodes and macroscopic algorithms deeply fusing, forcibly lock state vector evolution trajectories inside manifold $\Pi_{\perp}$, guaranteeing persistent anti-entropy persistence of complex entities on physical battlefields of factorial complexity $\mathcal{O}(N!)$.

Chapter 5: Mechanism: Decision Write-Back Mechanism of Control Domain Dimension $dim(C)$ and Mismatch Angle θ

5.1 Physical Essence of Decision Write-Back Mechanism (Human-Out-of-the-Loop)

Vol I Mechanism and Second Law (Persistence Axiom) noted: rules and mechanisms are work channels for systems to maintain persistence.

In system engineering, traditional management mechanisms often stay at "post-hoc recording, report dashboards, and manual meeting coordination". This open-loop mechanism physically introduces huge information time delays and subjective attenuation, unable to flatten dynamic impacts of factorial complexity $\mathcal{O}(N!)$.

Mechanism Governing Principle Established: Genuine control mechanisms are not post-hoc reports, but "human-out-of-the-loop, decision write-back". Optimal control vectors $u(t)$ solved by control system must be directly written back to physical entity nodes through automated channels (such as automatically rewriting execution node timing, autonomously triggering resource synergy). Dispensing with manual relay intervention is necessary and sufficient physical condition for mechanisms to achieve millisecond self-healing and anti-entropy persistence.

5.2 Alignment Mechanism Between Control Domain Dimension $dim(C)$ and Observation Domain Dimension

In mechanism control layers, control effectiveness of systems is strictly limited by underlying observation precision. According to cybernetic alignment equations, control domain dimension $\dim(C)$ is strictly limited by observation domain dimension $\dim(O)$:

$$\dim(C) \leq \dim(O)$$

When perceptual distortion or data fracture leads to observation domain shrinkage, control domain dimension forcibly collapses to zero ($\dim(C) = 0$), and control system falls into governance paralysis.

Alignment Mechanism Governing Principle Established: Prerequisite for keeping control domain open is maintaining lossless closed loop of holographic observation. Mechanisms, by continuously extracting perception residuals Δ, ensure control domain dimension $\dim(C)$ always covers orthogonal bases of state space, providing broad control freedom for automated write-back work.

5.3 Unlocking Multi-Head Consultation Deadlock and Self-Healing Write-Back Channel

In complex giant system decision mechanisms, since chief designers cannot be omniscient, control systems must sufficiently guarantee that various nodes and functional departments can **freely input objective data, field status, and demands** (free input).

However, **nodes cannot overstep authority to independently issue schemes**. According to social choice theory and cybernetics, if various nodes independently issue schemes and vote in committees, system inevitably falls into multi-head consultation deadlocks (Ashby, 1956) and governance paralysis.

Remedy of mechanism treatise lies in: **nodes freely input demands, Metasynthetic Engineering / Hall for Workshop centrally flattens conflicts and solves schemes**. Free input of nodes topologically represents holographic perception of actual reality residuals, while Metasynthetic Engineering central solving and write-back represents tensor work of silicon computation compensation and second-order amendment ($\Phi_{\text{Human}} \otimes \mathcal{A}_{\text{Silicon}}$). When external impacts cause system states to deviate from rigid trajectories, self-healing mechanism forcibly activates "human-out-of-the-loop" decision write-back channels; optimal control vectors u^* solved by Metasynthetic Engineering reverse-anchor (Reverse Causal Anchoring) along causal pointers to inject into microscopic entity nodes in milliseconds, forcibly re-routing traffic and resource allocation vectors, completely unlocking multi-head consultation deadlocks, driving system to seamlessly restore steady state.

Self-Healing Mechanism Governing Principle Established: Physical essence of mechanism work lies in completely eliminating multi-head consultation deadlocks through human-machine synergy tensor product topology of "node free input + central solving write-back", achieving millisecond self-healing re-routing of microscopic nodes based on reverse causal anchoring.

Chapter 6: Path: Dimension-Reduced Work Path from Microscopic Perception to Macroscopic Control

6.1 Microscopic Physical Perception Route (Filtering and Residual Extraction)

Vol I Path and three axioms noted: evolution of order must expand along deterministic logical paths. Path is spatiotemporal skeleton of dimension-reduced work.

Physical essence of dimension-reduced work path lies in holographic perception and filtering extraction of microscopic physical states.

On physical battlefields interwoven with sea of heterogeneous nodes, systems must never directly process disorderly raw noise. Primary link of path lies in filtering mapping physical phase space through prior boundary creation operator Π, extracting high-fidelity perception residuals $\Delta(t)$:

$$\Delta(t) = \Pi \cdot (x_{\text{real}}(t) - x_{\text{model}}(t))$$

Perception Path Governing Principle Established: Without accurate extraction of microscopic residuals, all subsequent deductions are water without source. Residual extraction completes key first step of converting physical gaps into logical algebraic scales.

> **[Reverse Backtracking Diagnosis Along Causal Topology Chain and Dual Alignment Verdict Theorem]**: After perception residual $\Delta(t) = x_{\text{real}}(t) - x_{\text{plan}}(t)$ is born, control engine does not conduct uncritical mean square error statistics, but **hangs residual back onto causal topology chain (Causal Pointer Chain) that originally generated it, investigating and judging backward along clues**:

1. **Verdict 1: Model correct, world shifted (First-order physical execution deviation / Local perturbation)**:
    - *Diagnosis*: Algebraic causal logic of causal topology chain is completely correct, only physical site underwent spontaneous perturbation and spatiotemporal shift;
    - *Control Action*: Axioms and models remain unchanged; algorithm converts residual into potential difference compensation for next clock cycle, pulling self-healing absorption inside boundary (first-order closed loop);
2. **Verdict 2: World correct, model bankrupted (Second-order cognitive collapse / Gödel residual $\Delta > \theta_{\text{trigger}}$)**:
    - *Diagnosis*: Residual exceeds critical threshold, and backtracking along causal topology chain reveals physical path is topologically blocked or axioms fail;
    - *Control Action*: Old causal rules and topological models completely bankrupt. Metasynthetic Engineering jumps outside loop to exercise constitutional amendment rights, activating second-order metacognitive constitutional amendment operator Φ: severing invalid causal pointers, reconstructing boundary axioms and network topology (second-order amendment). **World shifted, adjust execution; model wrong, second-order amendment.**

6.2 Interwoven Route of Parallel Simulation in Computational Space (Parallel Universe Pruning)

Hub link of path lies in concurrent parallel simulation and pruning inside computation space.

After residual $\Delta(t)$ is extracted, control system inputs data into evolution algorithm helix, concurrently simulating N sets of possible evolution trajectories (parallel universes) in computation space:

$$X_k' = \mathcal{F}(X, u_k), \quad k = 1,2,\dots,N$$

Pruning Path Governing Principle Established: Essence of algorithm path is absorbing risks in computation space. Algorithms, through rigid trajectory manifold $\Pi_{\perp}$, strip divergent infeasible branches, retaining only optimal control vectors u^* evolving along rigid manifold, completing dimension-reduction convergence from complex divergence to single orthogonal trajectory.

6.3 Write-Back Route of Macroscopic Physical Control (Human-Machine Dual Carriers and Uncontrollability Theorem)

Terminal link of path lies in write-back work and resource re-routing of macroscopic physical entity control.

Governance paths must simultaneously establish dual translation models of **"ought-to-be software system carrier" and "ought-to-be human/organization carrier"**. Cybernetic iron law notes: **"Unobservable means uncontrollable; system control must be built and mastered from controlled end (physical write-back end)"**.

Solved optimal control vectors u^* directly inject into physical entities along write-back channels, forcibly re-routing traffic and resource allocation vectors $f(t)$ of various nodes:

$$\sum f_{\text{in}}(t) = \sum f_{\text{out}}(t)$$

Write-Back Path Governing Principle Established: Final completion of path must take reverse work of controlled end and establishment of human-machine dual carriers as verdict criteria. Following mastery path of building backward from controlled end, system engineering successfully converts governance will 100% into anti-entropy order of entity domains, completing ultimate leap from qualitative description to work science.

Chapter 7: Dynamics: Newtonian Vector Work, Phase Interference, and the "Dual-Chief System" Supreme Work Contract

7.1 Newtonian Vector Work and Phase Interference Equation

Vol I Generation and Persistence Axioms established: maintenance and evolution of order must rely on continuous reverse work. Work is physical destination of dimension-reduced governance.

In open complex giant system engineering, various subsystems or management functional departments solve work vectors $\boldsymbol{v}_i$ under respective local goals. When lacking governing constraints of unified all-domain potential energy field and prior boundary creation operator Π, work vectors present obtuse angles or mutual counter-strikes in global phase space ($\exists \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle < 0$).

According to Newtonian mechanics and vector product deductions, effective work W_{eff} of system is strictly determined by total intent magnitude and cosine of mismatch angle θ:

$$W_{\text{eff}} = W_{\text{total}} \cdot \cos\theta$$

When mutual interference and friction exist among system sub-departments ($\theta > 0^\circ, \cos\theta < 1$), large amounts of effective work energy are physically "vector-cancelled", completely converting into internal friction waste heat of system:

$$\Delta W_{\text{heat}} = \sum_{i=1}^{N} \| \boldsymbol{v}_i \| - \left\| \sum_{i=1}^{N} \boldsymbol{v}_i \right\| \geq 0$$

Vector Work Governing Principle Established: Without orthogonal boundary creation and all-domain governance, joint work force tends to zero ($\sum \boldsymbol{v}_i \to 0$). If management interventions trigger phase angle interference ($\theta > 90°, \cos\theta < 0$), all system energy will completely dissipate into organizational friction and inventory sinking waste heat.

7.2 Elimination of Vector Cancellation and Convergence of Effective Work Conversion Rate

Physical essence of dimension-reduced work lies in forcibly establishing globally self-consistent scalar potential energy fields through prior boundary creation operator Π, stripping non-orthogonal degrees of freedom normally, eliminating work vector phase angle counter-strikes ($\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle \geq 0$).

When non-orthogonal internal friction degrees of freedom are forcibly stripped to zero ($x_\perp(t) \to 0$), mismatch angle is forcibly converged toward zero:

$$\theta \to 0° \Longrightarrow \cos\theta \to 1$$

At this moment, internal vector cancellation waste heat returns to zero ($\Delta W_{\text{heat}} \to 0$), and effective work conversion rate reaches physical limit:

$$W_{\text{eff}} = W_{\text{total}} \cdot \cos\theta \to W_{\text{total}}$$

Work Convergence Governing Principle Established: Prior boundary creation and rigid potential barriers are not human hegemony, but necessary and sufficient physical and cybernetic conditions for complex systems to break internal friction deadlocks, eliminate vector cancellation waste heat, and achieve 100% conversion convergence of effective work.

7.3 Qian Xuesen's "Dual-Chief System" Supreme Work Contract and Organizational Damping Absorption

To make a huge, heterogeneous open complex giant system with interest inertia work efficiently toward a unified goal, power endorsement must be physically introduced. Lacking supreme authorization endorsement, work vectors and management wills of subsystems will produce violent phase angle interference due to blurred power boundaries.

"Dual-Chief System" Metasynthetic Engineering contract proposed by Qian Xuesen established supreme physical mechanisms under Newtonian vector work perspective:

1. **Administrative Chief Line**: "Set direction, draw boundaries, give authorization", anchoring global scalar potential energy fields and ultimate attractors;
2. **Chief Designer Line**: "Build structure, obey physics, run closed-loop", responsible for solving five-dimensional dual helix and decision write-back.

Dual-Chief Contract Governing Principle Established: Administrative driving forces accurately focus on maximum resistance bottlenecks of physical bottlenecks (i.e., heaviest physical bottleneck impedance), eliminating phase angle interference of administrative and technical cross-boundary overstepping ($\theta \to 0°$), converting static friction damping of bureaucracy into co-directional work resonance, completing maximum efficiency convergence from organizational authority to physical work.

This physical reality demonstrates that classical laws of Newtonian mechanics on vector work and phase angle interference obtain rigorous causal continuation in complex system work governance, fully isomorphic and returned to positions in end-of-book star map.

Chapter 8: Evolution: Second-Order Metacognitive Constitutional Amendment Operator Φ and Generational Leap

> **[Living Evolution Volume Prelude Epigraph]**: "Open systems continuously exchange mass, energy, and information with external environment. According to Gödel's incompleteness theorem, finite boundary creation must be broken through by new residuals. Essence of living entities lies in relying on endogenous 'metacognition' to integrate new interference into system, completing dynamic re-boundary creation."

8.1 Absolute Conflict Between Intra-Boundary Local Optimization and Extra-Boundary Interaction Impact

Vol I Evolution Axiom and Third Law established: without residual feedback, generational evolution stagnates.

Systems inside established ought-to-be frameworks, through five-dimensional dual helix solving and vector work, reach intra-boundary local optimization (Boundary Creation Relativity Law: living is absolute, optimality is relative inside boundary creation). However, open complex giant systems physically are by no means isolated systems, must continuously exchange matter, energy, and information with extra-boundary disorder background.

When extra-boundary environment undergoes drastic paradigm phase transition or out-of-distribution (OOD) major impacts, extra-boundary non-linear perturbations forcibly penetrate boundaries, causing objective actual reality and intra-boundary expectations to produce violent residual divergence ($\Delta > \theta_{\text{trigger}}$).

8.2 Gödel Incompleteness Deadlock and Turing Halting Breakthrough

When residual divergence exceeds tolerance extreme of old frameworks, if system refuses to re-partition boundaries and foolishly attempts to prove or absorb new residuals within old rules and old axioms, first-order ought-to-be calculus inevitably falls into infinite loops of "explaining new phenomena using old definitions".

In mathematical logic and physical control, this state is strictly expressed as **Gödel Incompleteness Deadlock and Turing Halting** (Gödel Deadlock). First-order mechanical rules completely lose self-healing capability inside old boundaries.

8.3 Active Evolutionary Work of Second-Order Metacognitive Constitutional Amendment Operator Φ

Facing first-order Gödel deadlocks, systems can only seek survival by activating **Active State Second-Order Metacognitive Constitutional Amendment Operator** Φ. After perception residual $\Delta(t) = x_{\text{real}}(t) - x_{\text{plan}}(t)$ is captured, system forms dual alignment verdict topology: when residual belongs to local deviation, activate first-order closed loop inside boundary to adjust execution pulling

absorption; when residual exceeds critical threshold ($\Delta > \theta_{\text{trigger}}$) causing old axioms to bankrupt, system leaps to activate second-order metacognitive constitutional amendment operator Φ.

Operator Φ does not make local fine-tunings inside old frameworks, but observer (Chief Designer) jumps outside old boundaries, executing second-order metacognitive re-boundary creation work, executing constitutional amendment rewriting on underlying prior boundary creation operator Π and rigid manifold $\Pi_{\perp}$:

$$\Phi: \Pi_k \overset{\Delta X}{\to} \Pi_{k+1}$$

Second-order amendment operator Φ (evolutionary kinetic energy breaking deadlocks outward) and conscience compact support operator E_{supp} (safety potential energy converging boundaries inward) constitute Yin-Yang dual helix duality of system meta-governing layer: former ensures paradigms do not rigidify, latter ensures system never crosses generational survival red lines under extreme optimization.

Evolution Governing Principle Established: Evolution is not passive survival of fittest, but active constitutional amendment work expanded by observer with second-order metacognitive operator Φ, dual-core dually governed with compact support truncation of conscience operator E_{supp}. By breaking old dogmas through second-order re-boundary creation and integrating new dimensions inside new boundaries, system achieves generational leaps transcending Gödel deadlocks, reaching endless holographic anti-entropy persistence.

Volume IV: Industrial Validation and Decisive Data Verification

— Physical Verdict from the IPS Hundred-Billion-Scale Battlefield

Chapter 1: Phase Space Characteristics of the Verdict Battlefield

1.1 Factorial Complexity $\mathcal{O}(N!)$ of Massive Physical Execution Units and Kitting Constraints

On physical battlefields of global manufacturing networks, including LCFC Hefei (World Economic Forum “Lighthouse Factory”), system faces tens of thousands of physical execution units, heterogeneous multi-level kitting constraints, and dynamically fluctuating time limits.

Let physical node scale be N. Non-linear coupling and games between nodes trigger phase space combination possibilities to explode factorially $\mathcal{O}(N!)$. Facing this physical barrier, traditional reductionism analytical solutions completely diverge.

1.2 Physical Interpretation of Officially Verified Operational Performance Data

Validity of holographic anti-entropy theory and formal work operators obtained strict verification of official operational data in hundred-billion-scale discrete manufacturing networks:

1. **Delivery response rate rose from 54% to 98%**: With macroscopic self-sustenance of hundred-billion manufacturing network, qualitatively verified **Vol I Generation Axiom and high-fidelity filtering capability of prior boundary operator** Π (proving boundary creation effectively isolated background noise, observation domain fully opened to cover control domain);
2. **Order delivery accuracy increased by 32%**: Qualitatively verified **Vol I Persistence Axiom and normal stripping capability of rigid trajectory manifold** $\Pi_{\perp}$ (proving rigid potential barriers forcibly flattened non-orthogonal fluctuations, eliminating microscopic synergy friction);
3. **Overall inventory turnover increased by 1.9 times (releasing billions in liquidity)**: Qualitatively verified **Vol II Causal Finiteness Lemma 1.1 and closed-loop computability of decision automated write-back (**$T(N) \in \boldsymbol{P}$**)**, proving system completes non-friction convergence before physical instability, reducing sunk capital waste heat to minimum.

[Spatiotemporal Residual Three-Order Unified Theorem vs. IPS Empirical Data Physical Verdict Table]

Order	Physical Dimension	Core Control Law	Verdict Indicator	IPS Industrial Hard-Core Verified Data	Physical & Engineering Verdict Conclusion
First Order	Spatial Domain	Eliminate vector counter-strike inside boundary	**Structure does not fall apart**	Delivery response rate 54% → 98%	Proves prior boundary operator Π forcibly locks orthogonal bases, eliminating organizational and system phase angle interference ($\Delta W_{\text{heat}} \to 0$).
Second Order	Temporal Domain	Physical truncation of causal DAG depth	**Deduction does not time out**	Order delivery accuracy +32%	Proves causal depth truncation $K < +\infty$, single-step atomic operator polynomial solving ($T(N) \in \boldsymbol{P}$), deduction does not time out.

Order	Physical Dimension	Core Control Law	Verdict Indicator	IPS Industrial Hard-Core Verified Data	Physical & Engineering Verdict Conclusion
Third Order	Residual Domain	Gödel residual triggers Φ	**Paradigm does not rigidify**	Inventory turnover +1.9x	Proves second-order operator Φ reconstructs prior boundaries when heavy-tailed residuals touch extrema, eliminating capital sinking, paradigm does not rigidify.

[Qualitative Verdict of Industrial Battlefield on Four Falsifiability Deadlines]:

1. **Verification of Deadline 1 (Collapse without ought-to-be)**: Engineering practice confirms if confirmation of prior boundary operator Π is cancelled and five-dimensional ought-to-be model not built for heterogeneous elements, system identification matrix instantaneously diverges, and operational order physically collapses into breakdown;
2. **Verification of Deadline 2 (Dissipation without constraints)**: If normal constraints of rigid manifold $\Pi_{\perp}$ are removed, letting plants and departments game freely by local optima, vector phase angles severely counter-strike, friction dissipation surges, and system falls into paralysis and huge capital sinking;
3. **Verification of Deadline 3 (Stagnation without residuals)**: If residual feedback between actual execution and ought-to-be plans is cut off, system facing force majeure external phase transitions will stick to old plans, falling into Gödel incompleteness deadlocks, and evolution stagnates;
4. **Verification of Deadline 4 (Illusion of accessing actual reality without boundary creation)**: Any reductionist attempt trying to bypass ought-to-be medium to directly control full microscopic actual reality completely collapses facing factorial combination explosion $\mathcal{O}(N!)$. **Verdict Conclusion: Four falsifiability deadlines stand fully on physical battlefields; axioms of this system are self-consistent and rock-solid.**

Volume V: Cross-Domain Validation and Ultimate Evolution

— Cross-Domain Verdicts and Axiomatic Inheritance of Open Complex Giant Systems

Volume Prelude: Cross-Domain Verdict Criteria (Qualitative Eight Treatises vs. Quantitative Five Dimensions)

After completing axiomatic construction, work operator deduction, and decisive validation of hundred-billion industrial battlefields, this volume aims to establish cross-domain applicable boundaries of system and complexity science in cosmic all-domain phase space:

[Empirical Abstraction and Cross-Domain Verdict Logic Declaration]:

1. **Non-Quantitative Generalizability of Empirical Abstraction**: What hundred-billion industrial closed-loop empirical proof abstracted and established are **qualitative physical laws and cybernetic principles (i.e., "Qualitative Eight Treatises")** for self-consistent operation of open complex giant systems. This system does not attempt to directly provide unified microscopic quantitative algorithm schemes for heterogeneous cross-domain objects, because concrete quantitative computation physically and sufficiently relies on dimension-reduced modeling of five-dimensional models $\langle \mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X \rangle$ in specific entities of that domain.
2. **"Obey / Violate" Logic of Cross-Domain Verdicts**: Scientific logic of cross-domain cases (such as computational biology AlphaFold 2, modern cosmology black hole holographic entropy, macro-financial systems, silicon AGI, etc.) lies in **testing which principle treatises of "Eight Treatises" specific systems physically and cybernetically obey, or which treatises were violated/degenerated triggering systemic collapses**.
    - E.g.: AlphaFold 2 obeyed [Teleology] (boundary creation establishing ought-to-be boundary) and [Ontology] (silicon computation compensation performing tangent space algebraic pruning in phase space), thus successfully solving conformations;
    - Financial crises and Goodhart collapse originated from violating [Capability/Mechanism] (indicator arbitrage triggering control domain collapse $\dim(C) = 0$ and residual heavy-tail divergence);
    - Pure autoregressive AGI factual hallucinations and alignment drift originated from violating [Evolution] (closed first-order code lacking second-order metacognitive amendment operator Φ covering OOD phase transitions). **Any cross-domain system qualitatively satisfying "Eight Treatises" and three axiom constraints is automatically incorporated into this system in algebra, fully inheriting theorem guarantees of closed-loop computability ($T \in \boldsymbol{P}$) and extreme uniqueness ($\exists! S_k^*$).**

- **Qualitative Cross-Domain Verdict Criteria (Eight Treatises & Three Axioms)**: Whether a field (computational biology, black hole thermodynamics, financial systems, AGI) satisfies Generation Axiom (boundary creation establishing ought-to-be), Persistence Axiom (rigid rules establishing constraints), and Evolution Axiom (residual-driven evolution)? **If satisfying qualitative axiom conditions of Eight Treatises, it is automatically incorporated into this system, fully inheriting all physical and cybernetic properties of closed-loop computability ($T \in \boldsymbol{P}$) and extreme uniqueness ($\exists! S_k^*$).**
- **Quantitative Modeling Tools (Physical Five Dimensions $\langle \mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X \rangle$)**: Physical five dimensions are tools for conducting quantitative dimension-reduced modeling in specific cross-domain entities subsequently, rather than criteria for qualitative cross-domain verdicts.

Chapter 1: Cross-Domain Dimensionality Reduction and Axiomatic Interpretation

1.1 [Computational Biology] AlphaFold 2 and Physical Verdict on Levinthal's Paradox

Protein folding domain faces search dilemma of 10^{30} potential conformations. Success of AlphaFold 2 essentially lies in:

- **Verdict Conditions Satisfied**: Life molecules through boundary creation establish statistical barriers isolating internal/external perturbations (i.e. Markov blanket ought-to-be boundary, satisfying Generation Axiom), normally strip non-orthogonal thermal motion degrees of freedom along steepest descent trajectories (satisfying Persistence Axiom), extracting residuals to drive conformational leaps (satisfying Evolution Axiom).
- **[Qualitative Eight Treatises & Physical Law Isomorphism Verdict]**:
    - **Obeyed Treatises**: [Teleology] (boundary creation establishing ought-to-be conformation phase space), [Ontology] (silicon computation compensation executing tangent space algebraic pruning in phase space), [Scheme] (five-dimensional topological data model and evolution algorithm dual helix);
    - **Obeyed Laws**: Theorem 1.1 (Extreme uniqueness $\exists! S_k^*$), Work Waste Heat Stationary Value Minimization Principle.
- **Axiomatic Inheritance**: Satisfying qualitative Eight Treatises, protein folding is automatically incorporated into this system, its conformation solution converging to unique minimal internal friction geodesic ($\exists! S_k^*$), proving essence of AlphaFold 2 is executing algebraic pruning of phase space $\mathcal{O}(N!)$ by prior boundary operator Π in silicon computation space.

1.2 [Modern Cosmology] Black Hole Entropy, Holographic Principle, and Boundary Creation

Black hole event horizon is precisely the physical prior boundary creation operator Π of observer partitioning inside/outside phase space in curved spatiotemporal space.

According to Bekenstein-Hawking formula, black hole entropy is strictly proportional to 2D horizon area rather than 3D volume ($S_{\text{BH}} = \frac{k_B A}{4\ell_P^2}$). This classical result obtains profound formal interpretation under governing principles:

1. **Horizon as Boundary Operator** Π_{horizon}: Event horizon, as prior boundary operator seamlessly partitioning unobservable interior from external spatiotemporal phase space, determines that internal microstates can only project and count through 2D boundary quantized cross-sections;
2. **Cosmological Proof of Causal Finiteness Lemma 1.1 ($K < +\infty$)**: Since information transmission and work speeds in local spatiotemporal space are strictly limited by speed of light and Planck length ℓ_P, information degrees of freedom inside volume are normally stripped under gravitational collapse, making black hole maximum microstate count strictly limited by 2D boundary surface area A;
3. **Extreme Uniqueness and Holographic Isomorphism**: Horizon statistics of black hole entropy verify epistemological physical lower bound of Vol I "partition boundaries, ought-to-be manifests", proving Holographic Principle is essentially an inevitably derived corollary of prior boundary creation operator Π on curved extreme manifolds.

- **[Qualitative Eight Treatises & Physical Law Isomorphism Verdict]**:

- **Obeyed Treatises**: [Teleology] (event horizon as boundary creation operator Π_{horizon}), [Path] (partitioning boundary and reverse foundation from unobservable end);
- **Obeyed Laws**: Physical Causal Finiteness Lemma 1.1 ($K < +\infty$), Variational Principle Action Stationary Value Minimization.

1.3 [Macro-Financial Systems] Goodhart Collapse and Physical Truncation of Financial Crises

Financial systems, due to indicator arbitrage, trigger control domain collapse ($\dim(C) = 0$) and residual probability density $p(\Delta)$ heavy-tailed divergence (Goodhart collapse). Only by introducing moral risk compact support operator E_{supp} to forcibly truncate support set (satisfying Persistence and Evolution Axioms) can hacker arbitrage channels be cut off and global collapse flattened.

- **[Qualitative Eight Treatises & Physical Law Isomorphism Verdict]**:
  - **Violated/Degenerated Treatises**: Violated [Capability/Mechanism] (indicator arbitrage triggering control domain dimension collapse $\dim(C) = 0$), violated [Dynamics] (phase angle interference causing internal vector cancellation, $\sum \boldsymbol{v}_i \to 0$);
  - **Deadlock Unlocking Operator**: Introduce embodied conscience compact support operator E_{supp} truncating residual heavy-tailed distributions ($E_{\text{supp}} \cdot p(\Delta) \in$ Sub-Gaussian).

Chapter 2: Silicon Evolution: Second-Order Metacognitive Constitutional Amendment Operator Φ and Human-Out-of-the-Loop AGI Governance

2.1 Second-Order Metacognitive Constitutional Amendment Operator Φ and AGI Paradigm Rewriting

Current first-order AI architectures based on pure Transformer autoregressive language models and Scaling Law encounter severe dilemmas: facing out-of-distribution (OOD) complex environments, solely increasing parameter scale inevitably leads to phase space factorial explosion $\mathcal{O}(N!)$ and Goodhart collapse (such as alignment drift, hallucinations, and pseudo-self-consistency).

This system proposes hard-core formal remedies for AGI governance and evolution:

1. **First-Order Limitations and Residual Thresholds**: First-order AI systems are confined inside fixed prior boundaries Π_k. Facing out-of-distribution (OOD) environment impacts, when residual $\Delta > \theta_{\text{trigger}}$, first-order probabilistic models cannot reconstruct axioms independently;
2. **Activation of Second-Order Metacognitive Constitutional Amendment Operator Φ**: When residual exceeds trigger threshold, system must activate second-order metacognitive constitutional amendment operator $\Phi: \text{Boundary}_k \overset{\Delta X}{\to} \text{Boundary}_{k+1}$, rewriting underlying axiomatic bases and constraint clusters $\mathcal{C}$;
3. **Human-Out-of-the-Loop Governance and Meta-Governing Constraints**: Genuine silicon intelligence and safety governance lies not in soft Prompt fine-tuning or manual RLHF patches, but in encoding conscience compact support operator E_{supp} as an unbreakable "Second-Order Meta-Governing Principle" at AGI underlying layers. Systems during autonomous evolutionary leaps maintain vector work without phase angle cancellation ($\Delta W_{\text{heat}} \to 0$), achieving human-out-of-the-loop self-consistent safe operation.

- **[Qualitative Eight Treatises & Physical Law Isomorphism Verdict]**:
    - **Violated/Degenerated Treatises**: Violated [Evolution] (closed first-order autoregressive code lacking second-order metacognitive amendment operator Φ covering OOD phase transitions), violated [Ontology] (continuous token probability covering zero-measure rigid constraints, $\mu(C_{\text{feasible}}) = 0$);
    - **Deadlock Unlocking Operator**: Activate second-order metacognitive constitutional amendment operator Φ: $\text{Boundary}_k \xrightarrow{\Delta X} \text{Boundary}_{k+1}$.

Chapter 3: Panoramic Ultimate Isomorphism with Sages and Modern Science (Stars Returned to Their Positions)

Having traversed hundred-billion industrial empirical validation, five-dimensional dual helix solving, and four work operators, final volume of book looks back at human civilization puzzle over two thousand years: stars return to positions, streams return to source.

> **[Cross-Dimensional Isomorphism Ultimate Positioning Logic Declaration]**: Volume I Chapter 5 established human civilization's original inquiries on "boundary creation and order" in pure epistemological and philosophical dimensions. After completing axiomatic construction (Vol I), operator formalization (Vol II), dimension-reduced eight treatises (Vol III), and hundred-billion industrial empirical validation (Vol IV), this chapter as final volume of book completes panoramic ultimate isomorphism (stars returned to positions) of modern physics, cybernetics, system science, and behavioral intelligence masters in work science dimensions. Explain predecessors using axioms, rather than proving oneself using predecessors.

3.1 Modern Physics and Quantum Measurement: John Wheeler and Erwin Schrödinger

- **John Wheeler's "Participatory Universe"**: "It from bit" and participatory measurement are completely isomorphic with prior boundary creation operator Π—without observer's participatory boundary creation, physical reality in phase space is unobserved disorder.
- **Erwin Schrödinger's "Negative Entropy" and Variational Extrema**: Life "feeds on negative entropy"; classical variational principles (Fermat—Euler—Hamilton) prove adaptive systems rely on "boundary creation medium" to maintain steady state. Action stationary value minimization is connected to Persistence Axiom, and residual feedback is connected to generational evolution.

3.2 Classical Variational Principles and Information Theory: Fermat, Euler, Hamilton, and Claude Shannon

- **Fermat, Euler, and Hamilton's "Variational Principles and Extremal Internal Friction"**: Fermat's "least time path", Euler-variational "action stationary value", and Hamilton's "minimal internal friction trajectory" are strictly isomorphic with Persistence Axiom of this system: system evolves along rigid manifold trajectories $\Pi_{\perp}$ with minimal internal friction; variational stationary value minimization guarantees intra-boundary extreme uniqueness ($\exists! S_k^* \in \text{Sol}(\mathcal{C}_k)$).
- **Claude Shannon's "Information Entropy and Eliminating Uncertainty"**: Information entropy established by Shannon is strictly isomorphic with residual norm difference Δ of this system: information is elimination of uncertainty, and residual norm difference $\Delta = \| \Omega_t - \Omega_{t-1} \|$ is information gap between ought-to-be expectation and actual reality truth.

3.3 Cybernetics and Complexity Science: Norbert Wiener, W. Ross Ashby, and Qian Xuesen

- **Norbert Wiener's Cybernetics**: Core of cybernetics lies in reducing deviations through feedback; Wiener's deviation is prototype of residual norm difference Δ.
- **W. Ross Ashby's "Law of Requisite Variety"**: Only variety can absorb variety, isomorphic with rigid trajectory manifold $\Pi_{\perp}$ normally stripping redundant degrees of freedom and absorption capability.
- **Qian Xuesen's "Open Complex Giant Systems" and "Metasynthetic Method"**: Qian demarcated full-plane topology from simple systems to open complex giant systems; proposed Metasynthetic Engineering / Hall for Workshop thought of "combining human and machine, with human as primary" is formally expressed as tensor product work of human second-order metacognitive constitutional amendment operator Φ_{Human} and silicon first-order calculus algorithm $\mathcal{A}_{\text{Silicon}}$ ($\Phi_{\text{Human}} \otimes \mathcal{A}_{\text{Silicon}}$), landing as "Dual-Chief System" supreme work contracts on physical battlefields.

3.4 Pansystems Theory and Behavior Intelligence: Wu Xuemou and Cao Longbing

- **Prof. Wu Xuemou's "Pansystems Theory"**: Formally proved framework, relation, and transformation laws of complex systems in rigorous mathematics. "Pansystems boundary partitioning and transformation" is perfectly isomorphic with prior boundary creation operator Π performing algebraic pruning on phase space.
- **Prof. Cao Longbing's "Non-IID Learning"**: Revealed non-linear coupling essence of complex data and behavior dynamics. "Non-IID coupling" is isomorphic with non-orthogonal components of five-dimensional topological matrix A and emergence conditions of factorial complexity $\mathcal{O}(N!)$. This system provides formal work hands and human-out-of-the-loop self-healing mechanisms for pruning and reducing factorial complexity down to polynomial closed-loop computability ($T(N) = O(K \cdot N\log N) \in \boldsymbol{P}$).

When readers finish reading hundred-billion engineering empirical proofs, five-dimensional dual helix solutions, and work operators, looking back in this panoramic ultimate isomorphism exhibits true thought gravity of "explaining predecessors using axioms".

3.5 Panoramic Isomorphism Map of Ancient Sages and Modern Scientists

> **[Star Map Positioning Guide]**: This star map is not a list of ancient sages, but a closed-loop positioning map. Every ancient sage finds the position of wisdom fragments in this closed loop. When reading this map, do not look at rows, look at columns—every column is one facet of the closed loop.

Sage / Scientist	Core Insight	Corresponding Axiom / Operator in This System	Returned Volume & Chapter
Immanuel Kant	Human Mind Legislates Nature	Prior Boundary Creation Operator Π	Volume I, Chapter 5
George Spencer-Brown	Draw a Distinction	Axiom 0	Volume I, Chapter 5
Karl Popper	Falsifiability	Residual Norm Difference Δ Driving Evolution	Volume I, Chapter 5

Sage / Scientist	Core Insight	Corresponding Axiom / Operator in This System	Returned Volume & Chapter
Isaac Newton	Vector Work	Vector Work Equation & Phase Angle Interference	Volume III, Chapter 7
John Wheeler	Participatory Universe	Observer Emergence Law (Axiom 0)	Volume V, Chapter 3
Erwin Schrödinger	Feeding on Negative Entropy	Persistence Axiom	Volume V, Chapter 3
Fermat / Euler / Hamilton	Variational Extrema	Persistence Axiom & Stationary Value Minimization	Volume V, Chapter 3
Claude Shannon	Information Entropy	Residual Norm Difference Δ	Volume V, Chapter 3
Norbert Wiener	Feedback Reducing Deviation	Residual Norm Difference Δ	Volume V, Chapter 3
W. Ross Ashby	Requisite Variety	Rigid Trajectory Manifold $\Pi_{\perp}$	Volume V, Chapter 3
Qian Xuesen	Human-Machine Synergy	$\Phi_{\text{Human}} \otimes \mathcal{A}_{\text{Silicon}}$	Volume V, Chapter 3
Wu Xuemou	Pansystems Boundary & Transformation	Prior Boundary Creation Operator Π	Volume V, Chapter 3
Cao Longbing	Non-IID	Topology Matrix A Non-Orthogonal Coupling	Volume V, Chapter 3

Appendix and Corollary Set

Appendix A: Core Axiom Derivations and Quotations

Corollary A.1: Vector Cancellation Deadlock Theorem

In open complex giant systems lacking governance of prior boundary creation operator Π and constraints of rigid manifold $\Pi_{\perp}$, independent nodes blindly performing work driven by local interests cause intent vectors to produce non-orthogonal interference and mutual cancellation:

$$\left\| \sum_{i=1}^{N} \boldsymbol{v}_i \right\| \leq \sum_{i=1}^{N} \| \boldsymbol{v}_i \|$$

Sum of intent vectors of various nodes approaches zero. In control physics, this state is vector cancellation deadlock. System internal friction waste heat is maximized, effective work conversion rate returns to zero, and system inevitably collapses due to internal dissipation.

- **Axiomatic Conclusion**: Prior boundary creation and rigid potential barrier governing are not human hegemony, but necessary and sufficient physical and logical inevitabilities for complex systems to break vector cancellation deadlocks and establish negative entropy order.

Corollary A.2: Conscience Compact Support Operator and Second-Order Meta-Governing Principle Theorem (Goodhart Collapse Truncation)

When algorithm or system optimization pressure releases toward infinity, due to heavy-tailed residual distributions between proxy indicators and true intent, limit optimization inevitably causes expected utility of true intent to diverge to negative infinity:

$$\lim_{\text{optimization}\to\infty} E[r^*] = -\infty$$

Conscience compact support operator E_{supp} acts on residual probability density, forcibly restricting residual support set inside compact dead zone interval $[-\theta_{\text{dead}}, \theta_{\text{dead}}]$, truncating heavy-tailed distributions into light-tailed sub-Gaussian distributions.

- **Axiomatic Conclusion**: After truncation by conscience operator E_{supp}, true system utility expectations forcibly achieve analytical convergence, severing pathways of indicator arbitrage and system hacker collapse from mechanism level.

Corollary A.3: Physical Five-Dimensional Manifold Topological Invariant Theorem

Ought-to-be systems, during high-frequency information and energy exchanges with external actual reality environments, present highly non-linear emergence in external visual appearances, evolution trajectories, and work speeds. However, underlying orthogonal bases describing phase space states—Nodes, Topology, Constraint Clusters, State Vectors, State Transitions—are eternally invariant topological invariants.

- **Axiomatic Conclusion**: Missing any one of these five dimensions, ought-to-be system phase space tears mathematically in an instant, and system loses work capacity for anti-entropy exchange with external world.

Appendix B: Five-Dimensional Manifold Numerical Reconstruction and Algorithm Operation Manual

B.1 Operation Guide for Residual-Driven Parameter Numerical Modification of the Five-Dimensional Matrix

When perception residual Δ breaks evolution threshold triggering second-order metacognitive constitutional amendment operator Φ, algorithms cannot directly modify objective physical actual reality, but can only modify concrete parameter values (Values) in ought-to-be five-dimensional manifold matrix to achieve system reconstruction:

1. **Modify Node Parameters**: Update capacity upper limits, quota dead zones, open/close status, and operational attribute values of microscopic entities;
2. **Modify Topology Weight Values**: Topology is rules. Update weight values, logical routing, and interference coefficients of inter-node connection matrices;
3. **Modify Constraint Cluster Limit Values**: Constraints are boundaries. Update upper/lower limit dead zone values and resource quota limit values of rigid potential barriers $\Pi_{\perp}$;
4. **Modify State Vector Coordinate Values**: Update instantaneous measurement values of high-dimensional phase space coordinates of system;

5. **Modify State Transition Probability Values**: Algorithms are transitions. Update phase transition probability matrices, transition function step sizes, and evolution parameter values.

B.2 Formal Deduction of the Subjective Exclusivity of Second-Order Metacognitive Constitutional Amendment Operator Φ and Human Metacognition Theorem

1. **Gödel Incompleteness Deadlock of First-Order Silicon Computation**: Let first-order algorithm system be A, under constraints of given prior boundary axiom Π_k, its state calculus set is $S(\Pi_k)$. According to Gödel incompleteness and Turing halting deadlock, if residual $\Delta > \theta_{\text{trigger}}$ exceeding tolerance of Π_k exists, no provable and self-consistent mapping solution trajectory exists inside system Π_k, i.e., no $X \in S(\Pi_k)$ exists making residual $\Delta \to 0$.
    - **Conclusion 1**: Pure silicon first-order algorithms cannot complete self-rewriting of axiomatic basis Π_k inside own system.
2. **Physical Topology of Human Brain BA10 Area (Frontopolar Cortex) Metacognitive Evolution**: Evolutionary physics demonstrates human brain frontopolar cortex (BA10 area) expanded 4.7 times in relative volume compared to primates, evolving second-order self-referential awareness topology transcending first-order calculus. Second-order metacognitive constitutional amendment operator Φ is formally defined as ultra-formal mapping acting on prior boundary space: Φ:Map$(\Xi \to \Omega_k) \to$ Map$(\Xi \to \Omega_{k+1})$.
    - **Conclusion 2**: In known physical reality, human brain BA10 area is sole subject possessing physical entity structure of second-order metacognitive constitutional amendment operator Φ.
3. **Necessary and Sufficient Physical Condition Theorem of Human-Machine Synergy and Tensor Product Meaning**: Combining Conclusion 1 and Conclusion 2, overall work operator of open complex giant system completing dimension-reduced self-healing is expressed as:

$$\begin{aligned}&\text{System Work Operator}\\&= \text{Human Second-Order Metacognitive Constitutional Amendment Operator } \Phi_{\text{Human}}\\&\otimes \text{Silicon First-Order Evolution Algorithm } \mathcal{A}_{\text{Silicon}}\end{aligned}$$

Where ⊗ represents tensor product of unity of essence and function. Humans provide second-order amendment 'function' (Φ_{Human}), silicon provides first-order computation 'essence' ($\mathcal{A}_{\text{Silicon}}$); both spaces are orthogonally decoupled and indivisible, constituting a holographic tensor work system.

- **Theorem Established**: Human-machine synergy is necessary and sufficient physical condition for open complex giant systems to locally overcome computational irreducibility, achieving dimension-reduced self-healing and intra-boundary closed-loop computability.

Appendix C: Topological Isomorphism Map with Contemporary Frontiers

C.1 Isomorphism Table of "Diagnosis and Solution" with Prof. Longbing Cao's Non-IID Theory

- **Dimension 1: Reality Diagnosis**
    - Prof. Longbing Cao's Theory: Proving real data and behavior are Non-Independent and Identically Distributed (**Non-IID**) strongly coupled.

    - Monograph System: Proving non-orthogonal topological coupling exists between nodes, triggering factorial complexity $\mathcal{O}(N!)$.
    - **Physical & Cybernetic Isomorphic Conclusion**: Both sides from data science and theoretical physics perspectives jointly established objective existence of open complex giant systems.
- **Dimension 2: Solution Hands**
    - Prof. Longbing Cao's Theory: Extracting coupling features in high-dimensional space, conducting complex model fitting and feature classification.
    - Monograph System: Proposing prior boundary creation operator Π and rigid manifold $\Pi_{\perp}$ to execute algebraic pruning.
    - **Physical & Cybernetic Isomorphic Conclusion**: Monograph system provides dimension-reduced work hands for Non-IID theory to prune and reduce factorial complexity $\mathcal{O}(N!)$ down to polynomial complexity $O(K \cdot N\log N)$.
- **Dimension 3: Safety Defense Line**
    - Prof. Longbing Cao's Theory: Complex models under limit optimization face bias, hacker attacks, and collapse.
    - Monograph System: Proposing conscience compact support operator E_{supp} to forcibly truncate residual heavy-tailed distributions.
    - **Physical & Cybernetic Isomorphic Conclusion**: Injecting physical safety cut-off valves into Non-IID complex models to prevent Goodhart collapse.
- **Dimension 4: Ultimate Closed Loop**
    - Prof. Longbing Cao's Theory: Focuses on post-hoc data analysis, behavior prediction, and decision assistance (Human-in-the-Loop).
    - Monograph System: Establishing decision automated write-back and human-out-of-the-loop self-healing closed loops (Human-Out-of-the-Loop).
    - **Physical & Cybernetic Isomorphic Conclusion**: Helping Non-IID theory leap from post-hoc analysis to automated anti-entropy work closed loops with human out of the loop.

Appendix D: Full-System Keyword Master Cross-Reference Table ("One Table Linking the Whole Chain")

Cybernetic Work 5 Steps	Formal Work Operator (LaTeX)	Brain Mental Topology	Complex Giant System Physical 5D	Physical Work Essence & Logical Chain
1. Holographic Perception	**Residual Norm Difference** Δ	Hypersensitivity (AMY-LC) + Affect (SN)	Full Elements $\langle \mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X \rangle$	Holographically perceiving physical phase space full elements, filtering to extract high-fidelity perception residuals $\Delta(t)$, injecting evolutionary energy.

Cybernetic Work 5 Steps	Formal Work Operator (LaTeX)	Brain Mental Topology	Complex Giant System Physical 5D	Physical Work Essence & Logical Chain
2. Analysis	**Prior Boundary Creation Operator** Π	Cognition & Knowledge (Boundary Confirmation)	Topology Matrix A	$\Pi^2 = \Pi, \dim(\Omega) \ll \dim(\Xi)$. Locking orthogonal bases in topological structure, stripping non-orthogonal redundant degrees of freedom.
3. Decision	**Rigid Manifold** $\Pi_\perp$ / **Evolution Algorithm** $\mathcal{A}$	Rigid Potential Barrier (Potential) / Fluid Intelligence FPN (Kinetic)	Constraint Clusters $\boldsymbol{C}$ / State Transitions with Causal Pointers ΔX	Inside limits of rigid manifold $\Pi_\perp$, using algorithm $\mathcal{A}$ to concurrently simulate N sets of parallel universes in computation space, locking optimal dimension-reduced trajectories.
4. Execution	**Governing Work (Decision Write-Back)**	Holographic Mind Governing	Nodes N + State Vector X_{plan}	$W_{\text{eff}} = W_{\text{total}} \cdot \cos\theta$. Decision automated write-back injected into microscopic nodes, eliminating vector cancellation internal friction, driving angle $\theta \to 0$.
5. Feedback	**Second-Order Metacognitive Constitutional Amendment Operator** Φ	Pain Perception & Threshold Trigger	State Transitions with Causal Pointers ΔX	Capturing residual Δ. If $\Delta \leq \theta_{\text{trigger}}$, maintain first-order calculus; if $\Delta > \theta_{\text{trigger}}$, activate metacognitive constitutional amendment operator Φ to rewrite axiomatic bases.

Cybernetic Work 5 Steps	Formal Work Operator (LaTeX)	Brain Mental Topology	Complex Giant System Physical 5D	Physical Work Essence & Logical Chain
Top-Level Safety Valve	**Conscience Compact Support Operator** E_{supp}	Conscience (Default Mode Network DMN)	Constraint Cluster Dead Zone $[-\theta_{\text{dead}}, \theta_{\text{dead}}]$	$E_{\text{supp}} \cdot p(\Delta) \in$ Sub-Gaussian. Forcibly truncating heavy-tailed distributions, preventing algorithms under limit optimization from diverging to negative infinity (Goodhart collapse).

Appendix E: Core References

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